TECHNICAL REPORT

The Greek Poverty Paradox

Greeks report difficulty making ends meet at a rate far above what their official poverty rate would predict. This report works through the gap in eight stages, testing nine present-day constructs and eight accumulated ones against a pre-registered protocol, and reports what survived — including the substantial part that did not.

On the European Union's official measure of relative income poverty, Greek poverty is elevated but not exceptional. It ranks seventh of twenty-seven. On the European Union's official measure of subjective financial hardship — households reporting difficulty making ends meet — Greece ranks first, and has done so for years. The distance between the two runs to **52.6 percentage points** on average over 2015–2024.

That distance is the subject of this report. It could mean many things. It could be a measurement artefact, if the official poverty line is the wrong ruler. It could be that the broader official measure, AROPE, already covers it. It could mean Greek households are reporting something real that income poverty does not capture. It could mean Greeks answer surveys more pessimistically than other Europeans. Each of these is testable, and each is tested here.

The answer is layered rather than singular, and the honest version of it includes a large residue that the analysis could not resolve. That residue is reported as prominently as the findings.

Skip to the findings in one page

THE EIGHT STAGES

Findings appear as **indented statements with a coloured rule**, each followed by the limits that qualify it. Both are fixed — written down when the analysis finished, reproduced here word for word — and the limits matter as much as the findings themselves.

– How to read this document, and the evidence vocabulary

Those statements are fixed: they were written down when the analysis finished, before this report was drafted, and they are reproduced here word for word rather than paraphrased. The limits are fixed in the same way and matter as much as the findings -- several of them exist because an earlier draft of this report claimed more than the evidence carried.

Some material is discussed here without having been established here: institutional trust, migration, the adjustment programmes, tax incidence and the pre-crisis wellbeing comparison. It appears in **dashed boxes** that say what may and may not be concluded from it. Parts of it were examined -- migration was tested on this panel, and the cross-domain and ESS comparisons are descriptive analyses done for this report -- but none of it can carry a headline finding, and the boxes exist so that it is never mistaken for one.

Technical material elsewhere in the report sits behind expandable *methods* panels like this one. The main reading path does not depend on opening them; they are there so that every number can be traced to how it was produced.

The evidence vocabulary

Results are not reported as significant or not significant. A pre-registered decision rule, implemented as tested code rather than as prose, assigns each construct one of the following:

Supported

Cleared every pre-registered gate, including the wild cluster bootstrap.

Inconclusive under available power

Did not clear the gates, and the design could not have detected an effect of the relevant size. This is **not** evidence of absence.

Unsupported with adequate power

Did not clear the gates, and the design could have detected an effect of the stated magnitude. The exclusion is specific to that magnitude.

Blocked

Not testable without circularity, because the candidate shares an instrument with the outcome or overlaps it mechanically.

Infeasible

The data required does not exist. Distinct from a null result.

The distinction between the second and third of these does most of the work in this report, and collapsing them would misrepresent the findings substantially.

The findings, in one page

Everything below is set out in full over the eight stages that follow, with the evidence and the limits of each. A reader who stops here should still come away with an accurate picture.

52.6 percentage points between Greece's official income-poverty rate and the share of Greek households reporting difficulty making ends meet, averaged over 2015–2024. Greece ranks first of twenty-seven on the second measure and seventh on the first, and sits further from its nearest neighbour than that neighbour sits from most of Europe.

WHAT ACCOUNTS FOR PART OF IT

1. **The official measures are narrower than the experience.** The EU's broader measure closes about a fifth of the distance, and its contribution is shrinking. The income line itself fell with the Greek economy, which a relative measure cannot avoid doing.
2. **The reported difficulty corresponds to concrete failure.** It moves with arrears, unaffordable heating and unexpected expenses, within countries as well as across them — though every one of those items comes from the same survey as the question itself.
3. **Present conditions and accumulated history both carry information.** Material resources, long-term unemployment and wage-adjusted affordability each predict

WHAT THE EVIDENCE DOES NOT SUPPORT

1. **Nothing here shows what causes what.** Every result is a cross-country association across twenty-seven countries.
2. **Nothing here shows Greece changing over time.** The accumulated measures separate countries; the within-country tests are too imprecise either to establish a trend or to rule one out.
3. **Most measures that did not work are unresolved, not excluded.** Six of nine present-day measures are inconclusive because the study could not have detected an effect of the relevant size.
4. **One central result reverses.** Admitting a single same-survey predictor moves Greece from the third to the twenty-fifth

hardship beyond income poverty; so do accumulated unemployment, the years wages have stayed below 2008, and housing-cost deterioration, even after their present-day counterparts are accounted for.

largest unexplained gap in Europe, on identical data. Neither version is definitive.

And most of the gap is unexplained. Of the 52.6 points, what is accounted for above is a minority. The remainder is not attributed to anything here, because the analysis could not establish where it goes. A reporting tendency among Greek respondents cannot be excluded either, and a separate survey reaching back before 2008 suggests Greece started lower.

STAGE 1 **The puzzle**

Are the two measures really describing different things, and is Greece unusual or merely extreme?

The European Union measures poverty in two quite different ways, and both are official. The first, *at-risk-of-poverty* (AROP), counts households below 60% of their own country's median equivalised income. The second asks households directly whether they have difficulty making ends meet. The first is a position in a national income distribution; the second is a report about lived experience. There is no reason they must agree, and across most of Europe they broadly do.

In Greece they do not.

FIGURE 1 **Greece reports far more hardship than countries with similar income poverty**

DESCRIPTIVE Does official income poverty account for the hardship Greek households report?

Read with this. Descriptive comparison of two official measures. Reported hardship is answered by HOUSEHOLDS and income poverty counts PEOPLE, so the axis is labelled percent rather than either. In the second view the fitted line excludes Greece, describing the European pattern Greece is being judged against rather than one Greece helped set. The labelled point marks the median country on each measure taken SEPARATELY, so it is a reference point rather than an actual country: no member state necessarily sits there. Both views are country-level and say nothing about any individual household.

How Greece's hardship gap developed

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece: reported hardship	77.7	76.8	77.2	74.1	71.0	69.8	71.0	68.4	67.0	66.7
EU median: reported hardship	28.9	25.9	22.6	20.5	18.5	16.8	15.4	17.9	16.6	17.1
Greece: income poverty	21.4	21.2	20.2	18.5	17.9	17.7	19.6	18.8	18.9	19.6
EU median: income poverty	16.3	16.5	15.9	16.4	15.4	16.1	15.7	15.6	15.4	15.5

Where countries stood in 2024		
Series	**Income poverty (%)**	**Reported hardship (%)**
Czechia	9.5	14.2
Belgium	11.4	17.1
Denmark	11.6	12.8
Netherlands	12.1	7.3
Finland	12.6	8.9
Ireland	12.7	16.9
Slovenia	13.2	13.6
Poland	13.8	11.5
Austria	14.3	13.1
Hungary	14.3	19.8
Slovakia	14.5	28.7
Cyprus	14.6	20.8
Sweden	14.8	9.9
Germany	15.5	7.3
France	15.9	21.8
Portugal	16.6	21.8
Malta	16.9	17.3
Luxembourg	18.1	8.5
Italy	18.9	18.7
Romania	19.0	23.8
Greece	19.6	66.7
Spain	19.7	21.9
Estonia	20.2	9.9
Croatia	20.3	19.8
Lithuania	21.5	11.4
Latvia	21.6	23.3
Bulgaria	21.7	37.4

The two series never converge over the observed period. Reported hardship begins near 80% and remains above 60% throughout; income poverty moves within a narrow band around 20%. The distance narrows somewhat after 2016, which matters and is returned to in Stage 3, but it never closes to anything that could be called agreement. This is the finding the rest of the report is about.

> Greek subjective hardship runs 52.6 points above relative income poverty on average, and Greece ranks first of 27 on hardship while ranking seventh on AROP.

It is worth pausing on what a gap of this size means. Roughly one Greek household in five is counted as at risk of poverty. Roughly two in three report difficulty making ends meet. These are not two estimates of the same quantity that happen to differ; they are far enough apart that no plausible measurement error reconciles them. Either the official rate is missing most of what Greek households are experiencing, or Greek households are reporting something other than material difficulty. Both possibilities are taken seriously here, and the second is tested directly in Stages 3 and 7.

A word on the series itself. Eurostat's subjective-hardship indicator in its current form does not reach back before 2010, which would truncate the crisis at exactly the wrong moment. The earlier years shown here come from a constructed series, validated against the official one where the two overlap. That construction is ours, not Eurostat's, and it is flagged wherever it appears.

> The outcome is Eurostat's official subjective-hardship indicator, extended backwards before 2010 using a constructed series validated against it on 432 overlapping country-years.
>
> **Limits.** pre-2010 provenance is ours, not Eurostat's.

– How the pre-2010 series was constructed and checked

From 2010 onward the outcome is Eurostat's official indicator *ilc_sbjp01*, used directly and not recomputed. It reports the share of households declaring difficulty or great difficulty making ends meet. Before 2010 Eurostat does not publish it, so the earlier years are derived from the official components in *ilc_mdes09*, aggregated by the same rule that Eurostat's own series follows.

That aggregation rule was validated on **432 overlapping country-years** — every country and year where both exist. Of those, 318 agree exactly and 114 differ by a single rounding step; none differs by more than 0.1 percentage points, and the cross-country rank correlation is 1.00 in every year.

Two things this does *not* establish, and neither is claimed. It does not mean Eurostat published *ilc_sbjp01* before 2010; it did not. And it validates the **aggregation rule only** — whether the earlier national survey vintages are comparable across breaks is a separate question that this overlap cannot answer.

Two disciplines were imposed. First, the constructed values are used only for the descriptive picture in Stages 1 and 2; **every inferential test in Stages 4 through 6 runs on the official series alone**, so no statistical result depends on our construction. Second, the validation was run before the extension was used anywhere, not afterwards, and its result was recorded whether or not it was favourable.

The reason for extending at all is that a series beginning in 2010 begins after the Greek crisis was already underway. It cannot show a pre-crisis baseline, and a reader looking at it would see a plateau rather than a rise. The limits of this are real, and Stage 7 returns to a case where no comparable extension was possible.

Is Greece unusual, or just last?

Ranking first is not by itself remarkable — some country must rank first. The question is whether Greece sits at the end of a smooth distribution, in which case it is the extreme case of an ordinary European pattern, or whether it is detached from that distribution. The second view of the figure above answers it: fit the European relationship between income poverty and reported hardship through the other twenty-six countries, and Greece sits 47 percentage points above what that relationship predicts at its own income-poverty rate. The gap between Greece and the second-placed country is larger than the range spanning the middle of the distribution. Whatever produces this is not a stronger dose of what produces variation elsewhere. The full ranking is in the statistical appendix.

> **Where this leaves us.** Two official measures of the same underlying concept disagree by a wide margin for one country, and that country is separated from the rest of the distribution rather than continuous with it. The first thing to check is whether the European Union's own broader measure — which was designed precisely because income poverty alone was recognised as too narrow — already accounts for it.

STAGE 2 A broader measure, and a moving line

Does the EU's broader poverty measure close the gap, and did the yardstick itself move?

The European Union already accepts that income poverty alone is too narrow. Its headline social indicator is AROPE — at risk of poverty or social exclusion — which counts a household as affected if it falls below the income line *or* is severely materially deprived *or* lives in a household with very low work intensity. It is a deliberately broader net. If the puzzle in Stage 1 is simply that AROP is too narrow, AROPE should largely dissolve it.

FIGURE 2 **AROPE sits between reported hardship and income poverty, and closes only about a fifth of the distance**

DESCRIPTIVE What does AROPE add to AROP, and is that addition growing?

Read with this. AROPE components overlap and MAY NOT be added together: it is a union of income poverty, deprivation and low work intensity, and aggregate data do not reveal the overlaps The three measures are shown as they are reported, in shares of households, so the distance between them can be read directly. That distance is plotted on its own in the statistical appendix, where the points AROPE closes fall from 11.0 in 2015 to 7.3 in 2024.

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Reported hardship	77.7	76.8	77.2	74.1	71.0	69.8	71.0	68.4	67.0	66.7
AROPE	32.4	32.6	32.2	30.3	29.0	27.4	28.3	26.3	26.1	26.9
Income poverty	21.4	21.2	20.2	18.5	17.9	17.7	19.6	18.8	18.9	19.6

The three measures are shown as they are reported, in shares of households, so the distance between them can be read directly. AROPE sits where its design implies: above income poverty, because it counts more kinds of disadvantage, and far below what households themselves report. In 2024 income poverty stands at 19.6% of households, AROPE at 26.9%, and reported difficulty at 66.7%.

So it helps, and it is not enough. Adding deprivation and low work intensity to income poverty moves Greece about seven points closer to what its households report, out of a distance of forty-seven. More telling is the direction of travel: that contribution is *shrinking*, from eleven points at the start of the period to seven by the end. As a solution to this puzzle the broader measure is getting weaker rather than stronger, which is what the second view plots.

> Switching to AROPE closes 9.8 of those 52.6 points (19%), leaving 42.8 unexplained, and its contribution shrinks from 11.0 points in 2015 to 7.3 in 2024.

An aggregate can conceal as much as it reveals, and AROPE is built from three quite different conditions. Before concluding that the broader measure merely falls short, it is worth asking which of its parts moved, how each looks against the rest of Europe, and for whom.

The first three views take one measure each — income poverty, severe material deprivation, then AROPE itself — and show Greece against every other member state rather than against a single median. The last two split the Greek population by age and by sex.

FIGURE 3 **Where AROPE sits against the headline measures, what it's made of, and which groups moved**

DESCRIPTIVE

Where does AROPE sit against income poverty and reported hardship, what is AROPE made of, and did every age and sex group move the same way?

Read with this. These are changes in group-level rates, not evidence about the same individuals over time. The 2024-2025 national increase was driven primarily by within-group changes, especially deterioration among people aged 65+, rather than population ageing. AROPE components are a UNION and may not be summed. Very low work intensity is part of AROPE, but the project does not hold a comparable national-total series for the Components view; its available age coverage uses a different population base, and aggregate component rates cannot reconstruct the AROPE union in any case. In all four views colour marks the measure or group and dash marks the country, so each Greek line is paired with the EU median for the same thing in the same colour. The per-component views with every other member state drawn behind are in the statistical appendix.

Headline measures

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Income poverty: Greece	21.4	21.2	20.2	18.5	17.9	17.7	19.6	18.8	18.9	19.6
Income poverty: EU median	16.3	16.5	15.9	16.4	15.4	16.1	15.7	15.6	15.4	15.5
AROPE: Greece	32.4	32.6	32.2	30.3	29.0	27.4	28.3	26.3	26.1	26.9
AROPE: EU median	22.5	22.2	21.4	20.1	20.1	19.9	19.6	19.6	19.8	19.6
Reported hardship: Greece	77.7	76.8	77.2	74.1	71.0	69.8	71.0	68.4	67.0	66.7
Reported hardship: EU median	28.9	25.9	22.6	20.5	18.5	16.8	15.4	17.9	16.6	17.1

Components

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Income poverty: Greece	21.4	21.2	20.2	18.5	17.9	17.7	19.6	18.8	18.9	19.6
Income poverty: EU median	16.3	16.5	15.9	16.4	15.4	16.1	15.7	15.6	15.4	15.5
Material deprivation: Greece	17.6	18.4	18.3	16.1	15.8	14.9	13.9	13.9	13.5	14.0
Material deprivation: EU median	7.8	6.7	6.3	5.4	5.0	4.5	4.8	4.9	4.9	4.3
AROPE: Greece	32.4	32.6	32.2	30.3	29.0	27.4	28.3	26.3	26.1	26.9
AROPE: EU median	22.5	22.2	21.4	20.1	20.1	19.9	19.6	19.6	19.8	19.6

By age

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
All ages	32.4	32.6	32.2	30.3	29.0	27.4	28.3	26.3	26.1	26.9	27.5
Under 18	37.7	37.2	36.5	34.1	31.2	30.8	32.0	28.1	28.1	27.9	29.6
18-24	43.6	44.6	42.1	39.0	37.5	35.0	37.5	33.4	33.1	34.5	33.2
25-49	35.0	36.1	35.2	31.9	30.1	28.3	29.3	26.9	24.9	25.4	25.1
50-64	34.8	35.3	34.9	33.4	31.9	29.2	30.5	27.6	26.3	28.2	27.5
65 and over	17.5	16.6	18.4	19.2	20.5	19.4	19.3	21.0	23.9	24.9	27.7
EU: all ages	24.0	23.7	22.4	21.7	21.1	21.5	21.7	21.6	21.3	21.0	20.9
EU: under 18	27.4	27.1	25.1	23.9	22.8	24.1	24.5	24.8	24.9	24.2	24.3
EU: 18-24	29.7	29.8	28.1	27.2	26.6	27.8	27.3	26.5	26.0	26.3	26.3
EU: 25-49	23.3	22.8	21.3	20.3	19.3	19.6	20.2	20.0	19.6	19.2	19.1
EU: 50-64	25.6	24.4	23.3	22.4	22.1	21.5	21.9	21.0	20.8	20.7	20.8
EU: 65 and over	18.0	18.5	18.5	19.1	19.3	20.0	19.5	20.1	19.7	19.2	18.8

By sex

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
All	32.4	32.6	32.2	30.3	29.0	27.4	28.3	26.3	26.1	26.9	27.5
Women	33.1	33.4	32.9	31.3	30.1	28.6	29.2	27.4	27.3	27.9	29.1
Men	31.6	31.7	31.6	29.2	28.0	26.1	27.3	25.2	24.8	25.9	25.9
EU: all	24.0	23.7	22.4	21.7	21.1	21.5	21.7	21.6	21.3	21.0	20.9
EU: women	24.9	24.7	23.4	22.7	22.1	22.5	22.6	22.7	22.3	21.9	21.9
EU: men	23.1	22.5	21.4	20.6	20.0	20.4	20.7	20.4	20.3	20.0	19.8

The components do not move together, and the age groups do not move together either. This matters for interpretation: a single headline rate that combines a falling component with a rising one can look stable while the population underneath it is being reshaped. It does not close the gap, but it does explain why the aggregate looks quieter than the experience.

– What AROPE counts, and a coverage trap in the age breakdown

THE THREE COMPONENTS

A household is counted in AROPE if any one of three conditions holds. **At risk of poverty** means equivalised disposable income below 60% of the national median. **Severe material and social deprivation** means lacking at least seven of thirteen specified items, covering arrears, capacity to meet unexpected expenses, heating, diet, consumer durables and minimal social participation. **Very low work intensity** means working-age adults in the household used 20% or less of their combined work potential over the reference year.

Because the three are combined with OR rather than added, the headline rate is not the sum of its parts and cannot be decomposed into them without double-counting the overlap. The figure in this stage therefore shows the components separately rather than as a stacked total, and its first view is labelled for the two components it actually displays rather than implying all three.

A COVERAGE TRAP

The age-breakdown source does not carry a total row for every indicator. An earlier version of this figure intersected years across indicators and produced an empty default view — a chart that rendered, passed every structural check, and displayed nothing. The failure was invisible because the checks verified that the figure was well-formed, not that it contained data.

Every view of every figure is now required to carry at least two positions on its axis and at least one finite value. A chart can be perfectly well formed and still show nothing, and that is the kind of defect a reader notices at once while an automated check misses it entirely unless told to look.

LOW WORK INTENSITY AND THE AGE DIMENSION

Very low work intensity is defined only over working-age adults, so it behaves differently across age groups by construction: households composed entirely of people above working age are excluded from its denominator. Any comparison of AROPE across age groups is partly a comparison of which components can apply, and the figure separates the age groups rather than presenting a single aggregate that would hide this.

One further question sits behind the aggregate: when AROPE last rose, did that come from rates worsening inside age groups, or from the groups themselves changing size? The two can be told apart exactly rather than estimated, and the decomposition — charted in the statistical appendix — is unambiguous. It came from rates within groups; composition contributed almost nothing. A rise driven by an ageing population would be a different phenomenon from a rise driven by conditions deteriorating for people already counted.

> **Two different checks, not one.** AROPE broadens the *concept* of poverty: it counts more kinds of disadvantage. Anchored poverty changes the *yardstick*: it holds the income line fixed in real terms instead of letting it move with the national median. The first asks whether we are measuring enough things; the second asks whether the ruler itself moved. They are separate problems, and Greece has both.

What happened to the line

Relative income poverty counts people below 60% of the *current* national median. This is a deliberate design choice and a defensible one: it measures relative position, which is what it claims to measure. But it has a consequence that becomes severe in a large enough downturn. When national income collapses, the threshold falls with it. A household whose real income dropped sharply can remain above a line that dropped just as sharply, and will be recorded as not at risk of poverty throughout.

Greek median income fell by roughly a third over the crisis. The poverty line, being a fixed fraction of it, fell with it.

FIGURE 4 **Who counts as poor barely moved; what the poverty line buys fell by a fifth**

DESCRIPTIVE

What happened to the line Greek poverty is measured against, and what does a fixed line show instead?

Read with this. The anchored series is an approximation built in this project and is labelled as such wherever it appears. The first view counts PEOPLE, the second counts EUROS. In the second, both lines are the same official threshold: one as published in each year's own money, the other converted into what it could buy in 2008. The cash line recovers and the purchasing-power line does not, and that difference is why the first view's two counts diverge. Both views are Greece only and support no cross-country statement.

Who falls below a fixed line

Series	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece: fixed 2008 threshold	25.6	22.8	20.7	21.5	21.2	19.7	17.0	18.0	24.6	33.3	39.7	40.6	39.8	39.8	38.3	37.3	34.6	30.8	32.7	33.6	32.0	30.2
Greece: current-year threshold	20.7	19.9	19.6	20.5	20.3	20.1	19.7	20.1	21.4	23.1	23.1	22.1	21.4	21.2	20.2	18.5	17.9	17.7	19.6	18.8	18.9	19.6

What the line itself is worth

Series	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Threshold in cash terms	4923	5306	5650	5910	6120	6480	6897	7178	6591	5708	5023	4608	4512	4500	4560	4718	4917	5269	5251	5712	6030	6510	7020
Same threshold in 2008 purchasing power	5819	6089	6264	6343	6377	6480	6808	6768	6027	5168	4589	4270	4228	4216	4226	4338	4498	4884	4838	4815	4878	5113	5358

The two lines are the same threshold. One is the figure as published, in the euros of its own year; the other is that figure expressed in what it could buy in 2008. In cash the line has recovered almost entirely, from a 2010 peak of €7,178 to €7,020 in 2025, a fall of 2.2%. In purchasing power it has not: from €6,808 to €5,358, a fall of 21.3%.

A household is counted as poor or not poor against the first line. It lives against the second.

The second view asks the same question in people rather than euros: how many fall below a line held fixed at its 2008 real value, against how many fall below the line as it actually moved. This is not a criticism of AROP, which is doing exactly what it was designed to do. It is a statement about what AROP cannot register: a decline affecting the whole distribution at once leaves relative position largely undisturbed.

– Anchoring, equivalisation, and why the anchor year matters

Anchored poverty holds the income threshold at its real value in a chosen base year and carries it forward with consumer price inflation, rather than recomputing it from each year's median. The base year is therefore a consequential choice, and a badly chosen one can manufacture almost any result.

We use **2008**, the last year before the Greek downturn, and we fixed that choice before running the comparison. It was not selected by looking at which anchor produced the largest divergence.

> **This series is Greece-only.** It compares Greece against its own 2008 standard of living, and it contains no other country. It therefore supports no cross-country statement whatever: it cannot show that Greece's threshold fell further than anyone else's, only that it fell relative to where Greece itself started.

Incomes throughout are equivalised using the OECD-modified scale, which weights the first adult at 1.0, additional adults at 0.5 and children under 14 at 0.3. This is Eurostat's own convention and is applied identically to every country.

One limitation deserves emphasis. Anchored poverty and relative poverty answer different questions, and neither is the correct one in general. The relative measure asks whether a household has fallen behind its contemporaries; the anchored measure asks whether it has fallen behind a fixed standard of living. In a country where everyone became poorer together, these diverge sharply — and that divergence is the point, not a defect. A reader who wants a single number for Greek poverty will not find one here, and the recommendation in Stage 8 follows directly from that.

A further caution: the anchored series is descriptive. It enters no inferential test in this report, and no finding rests on it.

DESCRIPTIVE CORROBORATION

ANCHORED POVERTY, CORROBORATED BY INDEPENDENT MICRODATA

An independent study using Greek household microdata reaches a compatible estimate. Andriopoulou, Kanavitsa and Tsakloglou (2020) construct an anchored poverty rate on a

2007 base and find it reaches 48% at the 2013 income-year peak. This project's own anchored series, on a 2008 base, puts the comparable income year at 40.6%.

The two are not the same measurement. They differ in microdata source, anchor year, price-adjustment method and publication vintage, and neither was constructed with the other in mind. What they share is direction and rough scale: both find that holding the poverty line fixed reveals substantially more crisis-era deterioration than the relative rate shows.

What may be concluded. Independent microdata evidence points in the same direction: a fixed pre-crisis poverty line shows much larger crisis-era poverty than contemporaneous AROP.

Limitation. Treating 48% against 40.6% as a direct validation test, a bias estimate, or proof that this project's reconstruction is too low by seven to eight points -- the two differ in data, anchor year, price adjustment and vintage.

Source. Andriopoulou, E., Kanavitsa, E. & Tsakloglou, P. (2020), Decomposing Poverty in Hard Times: Greece 2007-2016. LSE GreeSE Paper No. 149. EU-SILC microdata via ELSTAT, analysed independently of this project. Anchored FGT0 poverty on a 2007 base reaches 48% at the 2013 income-year peak, against this project's own 40.6% for the same income year. source

One measure, or many?

A single detached measure invites a single explanation, and the most deflating one is that the measure is broken. So it is worth asking how much company that measure keeps. Take indicators of Greek economic and social conditions — wages, hours worked, prices, migration, savings, expectations — and for each one ask a deliberately crude question: is this country in the worst fifth of the Union? Then count, twice, on the same set.

FIGURE 5 **On the same 16 indicators, Greece went from the EU's worst fifth on 4 in 2008 to 11 in 2024**

DESCRIPTIVE Did Greek disadvantage deepen on a few measures, or spread across many?

Read with this. Descriptive summary of the condition, not a predictor of it. DESCRIPTIVE ONLY. Breadth was tested as a predictor of reported hardship in P3a and does not survive: on its own it is not significant (p = 0.12), and once the other accumulated measures enter the model its coefficient reverses sign. It summarises the condition rather than explaining it. THE BASKET IS FIXED: these are the 16 indicators with a valid EU position in both 2008 and 2024, chosen without reference to whether they improved or deteriorated, so the same things are counted at both ends and in every year between. Unemployment, youth unemployment and the employment rate are absent because comparable EU coverage for them begins in

2009; hours worked, pay per hour, the work-effort squeeze and real wages are all present, so this is not a basket without labour-market information. The outcome and every model covariate are excluded. THE EU-COUNTRY MEDIAN is the median of the 27 member states' own shares on this basket -- a median across countries, not Eurostat's population-weighted EU aggregate -- named accordingly rather than simply 'EU'. Croatia never reaches full 16-indicator coverage in any basket year and does not appear as a line for that reason. The 'Which measures' view's axis is POSITION, not value: the indicators have no common unit, so 0 is the best place in the Union to be on that indicator and 100 the worst, whichever direction 'worse' runs in.

Of 16 indicators, **4 were already in the EU's worst fifth in 2008, 7 entered it by 2024, and none left**.

Which measures

Series	2008 position	2024 position	Transition
Hours worked against hourly pay	59.3	100.0	entered worst fifth
Prices measured against wages	48.1	100.0	entered worst fifth
Pay per hour worked	55.6	100.0	entered worst fifth
Household income after inflation	51.9	100.0	entered worst fifth
Wages after inflation	51.9	100.0	entered worst fifth
The poverty line after inflation	51.9	100.0	entered worst fifth
Keeping the home warm	73.1	98.1	entered worst fifth
Hours worked each week	94.4	100.0	already worst fifth
What households expect of the year ahead	100.0	100.0	already worst fifth
Citizens leaving the country	86.4	96.2	already worst fifth
Income inequality	80.8	81.5	already worst fifth
Food prices	14.8	77.8	outside both years
Economic output per person	55.6	77.8	outside both years
Prices overall	53.7	70.4	outside both years
Housing and energy prices	63.0	59.3	outside both years
What households actually spend	48.1	48.1	outside both years

How many measures

Series	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece	25.0	18.8	43.8	50.0	50.0	50.0	43.8	50.0	50.0	62.5	56.2	56.2	56.2	56.2	56.2	62.5	68.8
EU-country median	6.2	12.5	9.4	12.5	18.8	18.8	15.6	12.5	18.8	18.8	12.5	12.5	18.8	12.5	12.5	18.8	12.5

The basket is fixed before the count is taken: every indicator with a valid EU position in both 2008 and 2024, chosen without reference to whether it improved or deteriorated. Sixteen qualify. On those same sixteen, Greece moved from the EU's worst fifth on four in 2008 to eleven in 2024 — 25.0% to 68.8% — and because the set is identical at both ends and in every year between, the rise cannot be an artefact of a growing denominator.

The "Which measures" view separates what changed from what was already true. Four indicators were in the worst fifth before the crisis and remain there; seven more entered it over the period; none left. Greece was not uniformly weak in 2008 and is not uniformly weak now — five of the sixteen sit outside the worst fifth at both ends. What happened is that seven further dimensions joined the four that were already there, and that is what the view shows: where each indicator moved. Sixteen of the twenty-five now place Greece at or near the bottom of the Union, and they are not variations on one theme: pay per hour, hours worked, the real value of the poverty line itself, household saving, what households expect of the coming year, and the number of citizens leaving the country all sit in the same place.

This is a description, not an explanation. We tested breadth as a predictor of reported hardship and it does not survive: on its own it is not significant ($p = 0.12$), and once the other accumulated measures are in the same model its coefficient reverses sign. A quantity whose sign depends on what else is in the model cannot carry an explanatory reading, so this figure summarises the condition rather than accounting for it. What it establishes here is narrower and still useful: the measure that put Greece at the top of Europe is not an isolated instrument behaving strangely. It sits inside a wide field of measures that moved with it.

> **Where this leaves us.** The broader official measure closes about a fifth of the gap and is closing less over time. The income line itself moved down with the economy, which the relative measure cannot register by design. Together these account for part of the distance but leave most of it standing. The next question is the uncomfortable one: if the official measures do not capture what Greek households are reporting, is what they are reporting real?

STAGE 3 Is the hardship real?

Does reported difficulty track concrete affordability failure, or does it float free of material circumstances?

This is the stage where the report could have ended. If Greek households report hardship without any corresponding material difficulty, then the puzzle is about how Greeks answer questions, not about Greek poverty, and the remaining stages would be measuring an artefact. So it has to be addressed before anything else is tested — though, as this stage shows, the available evidence narrows the question rather than closing it.

The check is whether reported difficulty moves with items that name concrete events rather than feelings: falling behind on bills, being unable to meet an unexpected expense, being unable to heat the home adequately, and severe material deprivation.

These are more specific than a general assessment of making ends meet, but they are not independent of it. Every one is *self-reported by the same household in the same EU-SILC interview*. A household inclined to answer the whole battery downbeat would move all of them together, so this test can show that reported hardship is coherent with concrete circumstances — it cannot show that it is validated from outside the instrument. That limitation runs through this entire stage.

FIGURE 6 Three affordability measures closely track hardship in Greece; falling behind on bills tracks it much less closely

DESCRIPTIVE CORROBORATION

Did concrete affordability difficulties rise and fall with what Greek households reported?

Read with this. Same-survey corroboration, not independent validation or causal evidence. Greek correlations with reported hardship: unexpected expenses 0.92 · material deprivation 0.94 · keeping the home warm 0.87 · falling behind on bills 0.37. Every line is a distance from its own 2015-2024 average, including the European median, so three series on different scales can share one axis: the shapes may be compared, the levels may not. The scale is the same in all four tabs. European median hardship is deliberately absent, since the question here is whether the concrete difficulty moved with Greek reports, not how Greece compares.

Unexpected expenses

Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece: reported hardship	5.7	4.8	5.2	2.1	-1.0	-2.2	-1.0	-3.6	-5.0	-5.3
Greece: unexpected expenses	4.7	4.9	4.0	1.7	-0.9	2.0	-2.4	-5.1	-4.4	-4.8
EU median: unexpected expenses	7.5	5.6	2.5	-0.2	-0.9	-1.8	-4.7	-1.1	-2.9	-3.6

Material deprivation

Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece: reported hardship	5.7	4.8	5.2	2.1	-1.0	-2.2	-1.0	-3.6	-5.0	-5.3
Greece: material deprivation	2.0	2.8	2.7	0.5	0.2	-0.7	-1.7	-1.7	-2.1	-1.6
EU median: material deprivation	2.3	1.2	0.8	-0.1	-0.5	-1.0	-0.7	-0.6	-0.6	-1.2

Keeping the home warm

Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece: reported hardship	5.7	4.8	5.2	2.1	-1.0	-2.2	-1.0	-3.6	-5.0	-5.3
Greece: keeping the home warm	7.6	7.5	4.1	1.1	-3.7	-4.5	-4.1	-2.9	-2.4	-2.6
EU median: keeping the home warm	1.5	-0.1	-0.2	-0.9	-0.6	-0.3	-1.1	0.8	1.1	-0.0

Falling behind on bills

Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece: reported hardship	5.7	4.8	5.2	2.1	-1.0	-2.2	-1.0	-3.6	-5.0	-5.3
Greece: falling behind on bills	5.8	4.4	1.4	-0.5	-2.1	-6.6	-7.1	2.0	3.8	-0.7
EU median: falling behind on bills	1.9	2.2	0.3	0.9	-0.1	-1.3	-1.2	-1.5	-0.5	-0.9

The first view is Greece alone, year by year. Each panel draws reported hardship and one affordability measure as distances from their own averages, so two series measured on different scales can share an axis. Three of them move with hardship closely: unexpected expenses at 0.92, deprivation at 0.94, inadequate heating at 0.87.

Falling behind on bills is the exception, and a sharp one: 0.37. Arrears require having credit and obligations to fall behind on, so a household that lost access to credit years ago, or never had it, can be in serious difficulty without ever registering. The item most often treated as the hard, factual anchor of financial distress is the one that tracks Greek hardship least closely.

Each tab also carries the European median for that same measure, so a reader can see whether the concrete difficulty was moving the same way elsewhere or only in Greece. And this is not a Greek peculiarity: pooled across all twenty-seven member states, once each country's own average is removed, the same relationships hold at 0.63 to 0.80. The binned version of that pooled comparison, and every observation behind it, are in the statistical appendix.

> Reported difficulty co-moves with arrears, inability to meet an unexpected expense, inadequate heating and severe deprivation at within-country correlations of 0.63 to

> 0.80.
>
> **Limits.** same-instrument corroboration, NOT independent validation; all items and the outcome come from EU-SILC; not uniform: arrears within Greece is 0.371.

The limits on that finding carry real weight. A household in a difficult position may answer the whole battery downbeat, and that alone would generate correlations of this size without any item independently confirming another. Nor is the pattern uniform: the within-Greece correlation for arrears is much weaker than the range headline suggests.

The arrears figure deserves particular attention because it is the item most often treated as the hard, objective anchor of financial distress. Within Greece it correlates with reported difficulty at 0.371 — well below the 0.63 floor of the headline range. Arrears depend on having credit and obligations to fall behind on, so a household that has already lost access to credit, or that never had it, can be in severe difficulty without registering arrears at all. A summary that reported only the range would imply a uniformity that the underlying items do not have.

A stronger version of the same check asks how much of Greece's anomaly these items can statistically absorb. Three numbers answer it.

SPECIFICATION	PERCENT OF HOUSEHOLDS
Prediction from income poverty and year alone	25
Prediction including four related survey items	58.2
What Greek households actually report	72

Greece, on the country-years where all four items and the outcome are observed. Both predictions are out-of-sample: the model is fitted without Greece.

> Those same four items statistically absorb 71% of Greece's baseline residual (+46.92 to +13.74).
>
> **Limits.** ABSORPTION, never explanation; shared instrument is not shared cause; diagnostic only; never a headline explanation.

The word *absorb* is doing precise work and is not a synonym for *explain*. Once concrete deprivation items are in the model, most of Greece's unexplained excess is no longer statistically distinguishable. That is a statement about shared variance, not about mechanism — and for the reason given above, part of that shared variance is the interview rather than the world. Stage 6 shows how much rests on admitting such a predictor at all.

– Within-country versus pooled correlation, and why the distinction matters

A pooled correlation computed across all country-years conflates two very different sources of variation: differences *between* countries, which are large and persistent, and movements *within* a country over time, which are what a claim about Greek households actually requires.

Between-country correlation is easy to generate and weak evidence. Richer countries have lower deprivation and lower reported hardship on essentially every measure, so almost any pair of welfare indicators will correlate strongly across countries without either telling us anything about mechanism.

The within-country figures reported here are computed by demeaning each series by its own country mean before correlating, so only deviations from each country's own average contribute. A country that is persistently high on both measures contributes nothing to a within correlation. This is why the within-country figures are reported in the claim and the pooled ones are not.

The heatmap in this stage displays both, side by side, precisely so the difference is visible rather than asserted. Several relationships that look strong pooled are considerably weaker within, and one changes sign. Any reader inclined to take the pooled panel as the finding should look at the within panel first.

On the absorption figure. The 71% reduction is the change in Greece's country residual when the four deprivation items are added to the baseline specification, on identical rows. It is not an R-squared, not a variance decomposition, and not a mediation estimate. Mediation analysis would require an identification strategy this design does not have, and none is claimed.

– How the convergence share is computed, and why it is dimensionless

The measures compared here are on incompatible scales: some are percentages of population, one is a purchasing-power figure in the thousands. An earlier version of this figure plotted a change of −2,819 purchasing-power standards on the same axis as a change of +48.8 percentage points, which is not a comparison at all — the visual impression was driven entirely by the choice of units.

The figure now shows, for each measure, the **share of its own 2015 Greece–EU gap that had closed by 2024**. Each measure is divided by its own starting gap, so the quantity is dimensionless and the measures are genuinely comparable: 1.0 means the gap closed entirely, 0 means it did not move, and a negative value means it widened.

Two limits follow from the construction. A measure whose 2015 gap was already small will show a large proportional movement for a small absolute one, so the share should be read alongside the underlying values, which the fallback table carries. And the share says nothing about which side moved: a gap can close because Greece improved, because the EU average deteriorated, or both. Decomposing that would require attributing movement to each side, which this figure does not do and which the caption states explicitly.

Convergence, and what it does and does not mean

Stage 1 noted that the gap narrows after 2016 without closing. Greece did not move in one direction across every measure behind it: some crisis-era conditions improved substantially, while household resources and affordability either recovered more slowly than the rest of Europe or deteriorated further. That matters because a falling unemployment rate alone cannot describe what households actually faced.

FIGURE 7 **Some gaps narrowed, especially long-term unemployment; wage, resource and affordability gaps widened**

DESCRIPTIVE Which measures converged toward the EU and which diverged?

Read with this. A NARROWING GAP DOES NOT MEAN GREECE IMPROVED. A gap can close because Greece caught up, because the rest of the EU deteriorated, or because everyone moved in the same direction at different speeds. This chart measures RELATIVE CONVERGENCE, not national recovery, which is why both endpoints for Greece and for the EU median are in the tooltip and the table. "71% of the gap closed" and "conditions improved by 71%" are entirely different claims. The plotted quantity is dimensionless because the underlying gaps are measured in percentage points, index points and PPS per head and cannot share an axis.

Measure	Share of 2015 gap closed	Greece 2015	Greece 2024	EU median 2015	EU median 2024	Gap 2015	Gap 2024
Long-term unemployment	0.71	16.40	5.40	3.60	1.70	12.80	3.70
Share below own GDP peak	0.55	24.84	11.20	0.00	0.00	24.84	11.20
Housing-cost overburden	0.40	45.50	28.90	8.70	6.70	36.80	22.20
AROPE	0.26	32.40	26.90	22.50	19.60	9.90	7.30
Income poverty (AROP)	0.20	21.40	19.60	16.30	15.50	5.10	4.10
Real poverty threshold	0.02	65.14	78.77	106.45	119.11	-41.31	-40.34
Material deprivation	0.01	17.60	14.00	7.80	4.30	9.80	9.70
Reported hardship	-0.02	77.70	66.70	28.90	17.10	48.80	49.60
Real household income	-0.06	67.79	84.13	102.25	120.59	-34.46	-36.46
Hardship gap against income poverty	-0.06	56.30	47.10	13.10	1.20	43.20	45.90
Hardship gap against AROPE	-0.08	45.30	39.80	6.40	-2.30	38.90	42.10
Real wages	-0.78	76.91	68.18	101.43	111.85	-24.52	-43.67
Material resources	-0.93	14769.70	21309.90	16230.40	24129.10	-1460.70	-2819.20
Wage-adjusted affordability	-2.55	141.89	173.09	127.22	120.97	14.67	52.12

Ten of the fourteen measures above, restated with the EU-country median spelled out in each measure's own unit and a plain-language reading.

MEASURE	GREECE, 2015 → 2024	EU-COUNTRY MEDIAN, 2015 → 2024	GREECE'S DISTANCE FROM
Long-term unemployment	16.4% → 5.4%	3.6% → 1.7%	
Share below own GDP peak	24.8% → 11.2%	0.0% → 0.0%	2
Housing-cost overburden	45.5% → 28.9%	8.7% → 6.7%	3
AROPE	32.4% → 26.9%	22.5% → 19.6%	
Income poverty	21.4% → 19.6%	16.3% → 15.5%	
Material deprivation	17.6% → 14.0%	7.8% → 4.3%	
Reported hardship	77.7% → 66.7%	28.9% → 17.1%	4
Real wages, 2008 = 100	76.9 → 68.2	101.4 → 111.8	24.5
Material resources	14,770 → 21,310 PPS	16,230 → 24,129 PPS	1,46
Wage-adjusted affordability	141.9 → 173.1	127.2 → 121.0	14.7

Distances retain each measure's original unit and are comparable over time within a row, not across rows.

The clearest improvement is long-term unemployment: Greece's rate fell from 16.4% to 5.4%, cutting its distance from the EU-country median from 12.8 to 3.7 percentage points. Housing-cost overburden also fell substantially. Both are genuine improvements, even though Greece did not fully close either gap.

The household-resource picture runs the other way. Real wages fell further below their 2008 level while the EU median rose over the same years. Material resources grew in Greece, but more slowly than the median, widening the PPS gap. Wage-adjusted affordability deteriorated sharply. Reported hardship itself fell, but the EU median fell by almost exactly as much, leaving Greece's distance from it essentially unchanged.

The pattern is not that Greece failed to recover on every measure. Labour-market and housing conditions improved substantially, but those gains were not matched by comparable recovery in wages, resources and purchasing power. This uneven recovery motivates the next stage's tests of which conditions remain associated with the hardship gap.

> **Limits.** This comparison is descriptive and does not identify a mechanism. A narrowing distance can reflect movement in Greece, movement in the EU-country median, or both; the endpoints show those movements but do not establish their causes. Distances retain each measure's original unit and should be compared over time within a row, not across rows. The appendix retains all fourteen measures, including real household income, the

> real poverty threshold and the two derived hardship-gap measures omitted here for space.

Whether these relationships hold across countries and within them is the question the correlation structure answers directly. The full matrices — within countries, between countries and pooled — are in the statistical appendix. The comparison that matters for this stage is narrower: what happens to each measure's relationship with reported hardship when the scope changes from between countries to within them.

FIGURE 8 **The real poverty threshold shows the only material sign reversal when the comparison moves within countries; the rest either hold their sign or start from a between-country correlation too close to zero for a reversal to be meaningful**

DESCRIPTIVE

Which relationships with reported hardship change when the comparison moves from between countries to within them?

Read with this. Correlations identify duplication and sign reversals. They do NOT select variables, and a reversal is a fact about the two scopes rather than evidence about mechanism.

Hardship correlations

Measure	Between countries	Within countries	Change
Real poverty threshold	0.095	-0.699	-0.794
Real wages	0.005	-0.654	-0.659
Real household income	-0.197	-0.755	-0.558
Share below own GDP peak	0.645	0.231	-0.413
Inflation	-0.047	-0.439	-0.393
Wage-adjusted affordability	0.709	0.525	-0.185
AROPE	0.617	0.772	0.155
Material deprivation	0.679	0.786	0.107
Long-term unemployment	0.788	0.689	-0.099
Housing-cost overburden	0.573	0.491	-0.081
Income poverty (AROP)	0.446	0.427	-0.019
Material resources	-0.587	-0.593	-0.006

Most relationships survive the change with their sign intact and their strength reduced. The real poverty threshold shows the only material reversal: it correlates weakly and positively across countries and strongly negatively within them. Real wages technically changes sign too, but its between-country correlation is +0.01, too close to zero for the reversal to mean anything. A reversal like that is a fact about the two scopes rather than evidence about mechanism, and it is the clearest illustration of why the within figures are the ones reported in the claim above.

Where this leaves us. The reported hardship corresponds to concrete affordability failure, so the puzzle is not simply an artefact of how Greeks answer surveys — though Stage 7 returns to the reporting-style question with better evidence than a correlation. What remains is to identify which conditions predict hardship beyond what income poverty already captures. That requires moving from description to inference, and from this point the report follows a pre-registered protocol.

STAGE 4 What current conditions explain

Which present-day conditions predict hardship beyond income poverty, and which merely could not be resolved?

Nine candidate constructs were specified before any of them was tested: material resources, labour-market exclusion, wage-adjusted affordability, housing costs, inflation, migration, and three others. Each was required to clear the same conditions, in the same order, and those conditions were fixed before any of them was tested.

The constructs themselves are worth naming, because two of them are composites whose behaviour is not obvious from their labels. **Material resources** is actual individual consumption per head in purchasing-power standards — what a household in a country can actually buy, rather than what it nominally earns. **Labour-market exclusion** is the long-term unemployment rate: people out of work for a year or more, which is a different quantity from unemployment overall and moves more slowly. **Wage-adjusted affordability** relates the price level a household faces to the wages available in its own country, so a country can score badly either by being expensive or by paying poorly. The remaining six cover housing costs, several inflation measures, migration and multi-domain breadth.

The gates matter more than the coefficients, so they are worth stating plainly. Each construct must add explanatory power beyond income poverty and year effects — it is not enough to correlate with hardship if AROP already captures it. It must survive multiplicity correction within its declared family. And it must survive a wild cluster bootstrap, which is far more demanding than a conventional standard error when the number of clusters is small.

FIGURE 9 **Three current-condition constructs survive the full testing sequence**

PRE-PLANNED CONFIRMATORY

Which current-level constructs survive every pre-registered condition, and which gate stopped the rest?

Read with this. Inconclusive is not evidence of absence. Bars are CLUSTER-ROBUST 95% confidence intervals, not bootstrap intervals: the wild cluster bootstrap determines the final support status, and two constructs whose intervals exclude zero here still fail it.

Construct	Effect (SD)	CI low	CI high	p	p FDR	Bootstrap p
Long-term unemployment	0.808	0.523	1.093	0.000	0.000	0.009
Material deprivation	0.720	0.239	1.202	0.003	—	—
Wage-adjusted affordability	0.687	0.386	0.988	0.000	0.000	0.001
Share below own GDP peak	0.620	0.343	0.897	0.000	0.000	0.401
Material resources	0.590	0.330	0.851	0.000	0.000	0.005
Housing-cost overburden	0.545	0.141	0.949	0.008	0.015	0.553
Real household income	0.315	-0.279	0.908	0.299	0.413	—
Inflation	-0.274	-0.817	0.268	0.321	0.413	—
Real wages	0.174	-0.501	0.850	0.613	0.653	—
Real poverty threshold	0.138	-0.464	0.740	0.653	0.653	—

Three constructs cleared every gate.

> Material resources predicts hardship beyond AROP and year effects (coef -0.0013, wild-cluster bootstrap p=0.0055).
>
> **Limits.** cross-country association; no causal claim.

> Labour-market exclusion predicts hardship beyond AROP and year effects (coef +4.3402, wild-cluster bootstrap p=0.0085).
>
> **Limits.** cross-country association; no causal claim.

> Wage-adjusted affordability predicts hardship beyond AROP and year effects (coef +0.3110, wild-cluster bootstrap p=0.0005).
>
> **Limits.** cross-country association; no causal claim.

> **Material resources and material deprivation are not the same thing, and the difference matters later.** Material resources is what a country actually consumes per head, adjusted for local prices, and it comes from national accounts and the purchasing-

power programme — measured independently of any household survey. Material deprivation is a survey item: the share of households reporting they cannot afford a specified list of things, collected in the same EU-SILC interview as the question about making ends meet.

So one is an outside measure of what money buys, and the other is the same households answering an adjacent question. That is why material resources can be a supported predictor here while material deprivation appears in this report only as corroboration in Stage 3 and as the contested predictor in Stage 6. Admitting the second into a model is the choice that moves Greece from third to twenty-fifth; admitting the first is not controversial at all.

Their directions are what one would expect and are worth stating, since a result that ran the other way would be a reason to distrust the measure rather than to report it. Higher material resources predict *less* reported hardship; more long-term unemployment predicts *more*; worse wage-adjusted affordability predicts *more*. Each direction was declared before testing, and a construct whose coefficient had pointed the other way would have been recorded as contradicting its declared direction rather than quietly reinterpreted.

These are cross-country associations. Nothing in this design supports a causal reading, and the limits attached to each say so. What they establish is that a country's material resources, its stock of long-term unemployment and its wage-adjusted affordability each carry information about reported hardship that the official income-poverty rate does not.

FIGURE 10 **What the three supported constructs look like**

PRE-PLANNED CONFIRMATORY

What do the three supported constructs actually look like for Greece against the EU?

Long-term unemployment

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece	16.4	15.4	14.3	12.5	11.3	10.5	9.2	7.7	6.2	5.4
EU median	3.6	3.1	2.7	2.1	1.9	1.8	2.0	1.9	1.7	1.7

Material resources

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece	14770	14786	15372	15833	16372	15475	16772	19402	20701	21310
EU median	16230	16494	17038	18006	18707	17386	19479	21317	23055	24129

Wage-adjusted affordability

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece	141.9	148.6	148.2	156.8	162.2	163.3	165.6	171.2	173.1	173.1
EU median	127.2	124.7	124.6	119.8	118.4	117.7	120.2	120.7	122.3	121.0

DESCRIPTIVE CORROBORATION

INDEPENDENT DESCRIPTIVE CORROBORATION (GREECE IN FIGURES)

An independently published analysis on the Greece in Figures site reaches a similar descriptive picture from Eurostat and ELSTAT releases published in 2025 and 2026, without access to this project's panel or its testing protocol: Greece is not the EU's poorest country on actual consumption per head, but Greek employees work among the longest hours in the Union for comparatively low hourly reward, and everyday categories such as food and information and communication are expensive relative to what that pay buys. That is the same shape as material resources and wage-adjusted affordability above, reached by a different route and a different, more recent, vintage of data.

The article's central numbers reproduce against the primary releases they draw on, with two exceptions worth flagging on their own terms rather than folded into this project's findings: a stock of registered vehicles is presented as evidence of new-car purchases, which it cannot support, and a rise in resident trip-taking is generalised to "all Greeks travelled," which the underlying survey does not say. Three of its comparisons sit close to but not exactly on this project's own numbers — an AIC rank, an annual-hours measure against this report's weekly-hours one, and an overall price-level index — and each gap traces to a different year vintage or a different aggregate, not to a disagreement about the underlying facts.

What may be concluded. The independent picture -- Greece not last on actual consumption, long hours for comparatively low hourly reward, high relative prices in food and information/communication -- is consistent with and external corroboration for this project's own material-resources and wage-adjusted-affordability findings. Its central descriptive numbers reproduce against the primary Eurostat and ELSTAT releases they draw on.

Limitation. Treating the article's own comparisons, or its constructed AIC-per-hour ratio, as inferential evidence: it applies no bootstrap, no FDR correction, no within/between decomposition and no model comparison, and it does not test whether these factors account for reported hardship. Merging its 2025/2026-vintage figures into this project's frozen 2015-2024 panel. Repeating its car-ownership sentence, which cites a vehicle-STOCK figure as evidence of new purchases, or its 'all Greeks travelled' generalisation, which the underlying ELSTAT survey does not support. Reading its AIC rank, hours rank or price-level figure as contradicting this project's own numbers without first noting the different vintage or aggregate behind each.

Source. Greece in Figures, "Γιατί οι Έλληνες νιώθουν τόσο φτωχοί" [Why Greeks feel so poor]. Constructs an AIC-per-hour ratio from aggregate actual individual consumption and aggregate annual hours worked; cites 2025-vintage Eurostat AIC and price-level releases and a 2026 ELSTAT resident travel bulletin. source

The six that did not, and why that is not a null result

Six of the nine did not clear the gates. It would be convenient to describe these as ruled out. That would be wrong for most of them, and the distinction is the most important methodological point in this report.

> Six of nine current-level constructs are inconclusive under available power, not unsupported; two cleared FDR and collapsed under the bootstrap (p=0.40 and 0.55).
>
> **Limits.** inconclusive is not evidence of absence.

A test that fails to detect an effect tells us something only if it could have detected one. With twenty-seven countries and roughly a decade of observations, the design has limited power, and for most of the six constructs the smallest effect it could reliably detect is larger than any effect worth caring about. Those constructs are *inconclusive under available power*: the study is silent about them, not negative.

A small number are genuinely different. Where the design had adequate power and still found nothing, the exclusion is real — but it holds at a stated size, not in general. The inflation measures are the clearest case: annual food and housing inflation can be excluded at the magnitude this design could detect, which is not the same as showing that inflation does not matter to Greek households.

> Annual food and housing inflation are unsupported with adequate power at 0.70 SD; annual headline inflation is unsupported at its detectable conditional magnitude; compounded inflation since 2008 remains inconclusive.
>
> **Limits.** the exclusions are narrow and magnitude-specific; compounded inflation is INCONCLUSIVE, not ruled out.

T1 **All nine current-level constructs, with the gate that stopped each one that did not clear.**

CONSTRUCT	COEFFICIENT	FDR P	BOOTSTRAP P	VERDICT	STOPPED AT
Proximate material hardship	+1.4524	—	—	**Blocked**	proximity to outcome
Loss against own past Real wages	-0.0924	0.6531	—	**Inconclusive**	insufficient power
Loss against own past Real household income	-0.2168	0.4131	—	**Inconclusive**	insufficient power
Loss against own past Real poverty threshold	-0.0556	0.6531	—	**Inconclusive**	insufficient power
Loss against own past Share below own GDP peak	+1.8675	<0.0001	0.4005	**Inconclusive**	wild cluster bootstrap
Inflation exposure	-0.9925	0.4131	—	**Inconclusive**	insufficient power
Housing pressure	+1.1385	0.0147	0.5530	**Inconclusive**	wild cluster bootstrap
Material resources	-0.0013	<0.0001	0.0055	**Supported**	—
Labour-market exclusion	+4.3402	<0.0001	0.0085	**Supported**	—
Wage-adjusted affordability	+0.3110	<0.0001	0.0005	**Supported**	—

Coefficients are on each construct's own scale and are not comparable across rows. Verdicts are the output of the pre-registered decision function, not a reading of the p-values.

Two constructs illustrate why the bootstrap gate was worth imposing. Both cleared multiplicity correction comfortably and then collapsed under the bootstrap, with p-values of 0.40 and 0.55. Had the protocol stopped at the conventional step, both would have been reported as findings.

– The decision rule, the bootstrap, multiplicity, and minimum detectable effects

HOW A VERDICT IS REACHED

Each construct is passed through the same conditions in the same order, and the first condition it fails is the one recorded against it in the table above. The verdict is the output of that procedure rather than a reading of the numbers, which is what makes "inconclusive" and "unsupported" separable at all: they are different exit points, not different degrees of the same judgement.

THE WILD CLUSTER BOOTSTRAP

Standard errors clustered on twenty-seven countries are unreliable: cluster-robust inference is asymptotic in the *number of clusters*, and twenty-seven is not large. The wild cluster bootstrap addresses this by resampling cluster-level weights rather than observations.

The implementation imposes the null — the restricted, rather than unrestricted, variant. This is not a detail. An earlier version resampled from the unrestricted residuals and produced a p-value of 0.82 for a coefficient with a t-statistic of 9.69, which is not a marginal disagreement but a symptom of a misspecified bootstrap. Under the null-imposed version the same coefficient behaves as its t-statistic implies. Every bootstrap p-value in this report uses 1,999 replications with the null imposed.

MULTIPLICITY

Testing nine constructs invites false positives. Benjamini-Hochberg false-discovery-rate control is applied **within each declared family**, where the families were declared before testing. Applying FDR across all tests regardless of family would be more conservative but would also be incoherent, since the families ask different questions. The families were fixed before testing; reassigning them afterwards would invalidate the correction.

MINIMUM DETECTABLE EFFECTS

For every construct that did not clear the gates, the minimum detectable effect was computed *by simulation* rather than by formula: the observed panel structure is retained, an effect of known size is injected, and the proportion of simulated datasets in which the protocol recovers it is recorded. The MDE is the smallest injected effect recovered at least 80% of the time.

This is what licenses the distinction between inconclusive and unsupported. A construct is only ever described as unsupported when its MDE is smaller than an effect size that would matter substantively, and the report states the magnitude at which the exclusion holds. For

Stage 5 the MDEs are computed pair-specifically and conditionally, because the relevant question there is whether an accumulated measure adds anything *given* its current-level counterpart, which is a different and harder test.

LEAVING ONE COUNTRY OUT

Twenty-seven countries is few enough that a single one can carry a result. Every construct is therefore refitted twenty-seven times, each time with one country removed, and the range of coefficients across those refits is recorded alongside the headline estimate.

The three supported constructs are stable under this: long-term unemployment ranges from 3.20 to 4.83 against a headline of 4.34, wage-adjusted affordability from 0.25 to 0.33 against 0.31, and material resources holds its sign and magnitude throughout. None of them depends on any single country, and in particular none depends on Greece — which matters, because Greece is the case the report is about and a result driven by it would be circular.

Housing pressure behaves differently. Its refits run from −0.10 to 1.27, so dropping one country flips the sign. That instability is one of the reasons it does not clear the gates, and it is a more informative fact about that construct than its p-value.

WHAT THE SPECIFICATION CONTROLS

All models include the country's AROP rate and year fixed effects. Year effects absorb Europe-wide shocks — the financial crisis, the pandemic, the 2022 energy shock — that would otherwise be attributed to whichever construct happens to move at the same time. Controlling for AROP is what makes this a test of incremental information: a construct that merely proxies income poverty cannot clear the first gate.

Where this leaves us. Three present-day conditions carry information beyond income poverty; six are mostly unresolved rather than excluded. But every one of these measures describes Greece *now*. A country that has been in difficulty for a decade may differ from one with identical current conditions and no such history. That is the next question, and it is the one where overstatement is easiest.

STAGE 5

What accumulated history adds

Does the length of a country's difficulty carry information beyond its present state?

Two countries can look identical today and have arrived there by different routes. One has had 8% long-term unemployment for a year; the other has had it for twelve. The intuition that these are not the same situation is strong, and it is the intuition behind every account of the Greek crisis that emphasises its duration rather than its depth at any single moment.

Intuition is not evidence, and this stage is where an appealing story is easiest to overstate. What follows is stated narrowly on purpose. Three readings of the results were caught and corrected during review, each of them a version of the same error: treating a cross-country association as though it described a process unfolding within Greece over time.

Building the accumulated measures

For each construct supported in Stage 4, a matching accumulated measure was built: the total excess exposure a country has absorbed since a fixed baseline, rather than its level today. Excess unemployment accumulates the amount by which a country's rate exceeded its own pre-crisis norm, summed over years. Wage non-recovery counts the consecutive years real wages have remained below their 2008 level. Housing deterioration accumulates the worsening in housing cost burden since 2010.

FIGURE 11 **Greece ranks first on accumulated unemployment and housing deterioration, and second on wage non-recovery**

POST-SELECTION ROBUSTNESS

How much accumulated unemployment has each country absorbed, and where does Greece sit?

Read with this. cross-country association | no demonstrated within-Greece dynamic | no causal claim | current conditions not ruled out The three measures are in different units and may not be compared with each other: accumulated unemployment and housing deterioration are percentage-point-years, wage non-recovery is a count of consecutive years.

Accumulated unemployment

Country	Accumulated unemployment (percentage-point-years above 2009)
Greece	137.5
Cyprus	64.1
Croatia	43.5
Spain	33.1
Italy	32.9
Bulgaria	26.6
Portugal	20.6
Slovenia	20.6
Netherlands	13.1
Luxembourg	9.1
Ireland	8.9
Slovakia	8.9
Poland	7.7
France	5.9
Lithuania	5.6
Denmark	5.6
Hungary	3.2
Estonia	3.1
Finland	2.9
Austria	2.5
Belgium	2.4
Romania	2.4
Latvia	2.0
Czechia	1.2
Sweden	0.6
Malta	0.0
Germany	0.0

Years wages below 2008

Country	Years wages below 2008 (consecutive years below the 2008 level)
Hungary	16.0
Greece	15.0
Italy	14.0
Finland	3.0
Sweden	2.0
Slovenia	0.0
Romania	0.0
Portugal	0.0
Poland	0.0
Netherlands	0.0
Malta	0.0
Latvia	0.0
Luxembourg	0.0
Lithuania	0.0
Austria	0.0
Ireland	0.0
Belgium	0.0
Croatia	0.0
France	0.0
Spain	0.0
Estonia	0.0
Denmark	0.0
Germany	0.0
Czechia	0.0
Cyprus	0.0
Bulgaria	0.0
Slovakia	0.0

Country	Housing deterioration since 2010 (percentage-point-years above 2010)
Greece	233.2
Bulgaria	116.3
Luxembourg	37.8
Portugal	35.8
Sweden	17.1
Estonia	11.8
Slovenia	9.9
Hungary	9.6
Germany	9.4
Italy	8.2
Belgium	7.9
Finland	7.0
France	6.6
Slovakia	6.1
Latvia	5.7
Netherlands	4.9
Ireland	4.6
Spain	4.5
Malta	4.5
Austria	4.3
Poland	4.2
Romania	4.2
Czechia	3.8
Cyprus	2.1
Lithuania	0.5
Croatia	0.0
Denmark	0.0

Housing deterioration since 2010

Greece has absorbed a large accumulated exposure on these measures. On one of the three it is not the highest — Hungary has a longer run of wage non-recovery — and the figure shows the full ranking rather than Greece alone, so that this is visible rather than buried.

The test that matters

Showing that Greece accumulated a lot is description, not inference. The question is whether an accumulated measure predicts hardship *once its own current-level counterpart is already in the model*. If today's unemployment rate is controlled for, does the history of unemployment still add anything? That is a demanding test, because the two are strongly correlated by construction.

FIGURE 12 **Accumulated history provides additional conditional cross-country information in three of eight pairs**

POST-SELECTION ROBUSTNESS

Does accumulated history add anything once today's conditions are in the same model?

Read with this. cross-country association | no demonstrated within-Greece dynamic | no causal claim | current conditions not ruled out This shows the pairs that survive; all eight, including the five that do not, are in the statistical appendix. The axis is STANDARDISED: SDs of hardship per SD of the predictor. The eight measures are in different units - percentage-point-years, years, index points, percentages - so raw coefficients could not share an axis, and the raw value is given in the table and the tooltip instead. Each coefficient is tested against ITS OWN pair-specific detectable effect, not a study-wide threshold: conditional power depends on how much independent variation survives once the counterpart is controlled. Bars are cluster-robust intervals; the bootstrap decides support.

Test	Standardised effect (SD)	Raw coefficient	p FDR	Bootstrap p	Conditional MDE (SD)
Accumulated unemployment (controlling for Long-term unemployment)	0.550	0.294	0.000	0.003	0.600
Years wages below 2008 (controlling for Real wages)	0.624	2.090	0.005	0.006	0.500
Housing deterioration since 2010 (controlling for Housing-cost overburden)	0.668	0.242	0.000	0.002	0.600

Three accumulated measures cleared it.

> Accumulated excess unemployment predicts hardship (bootstrap p=0.0025) and retains a cross-country association after its current-level counterpart is controlled (conditional bootstrap p=0.0025).
>
> **Limits.** cross-country association; no demonstrated within-Greece dynamic; no causal claim; current conditions not ruled out.

> Duration of real wages below their 2008 level predicts hardship (bootstrap p=0.0245) and retains a cross-country association after its current-level counterpart is controlled (conditional bootstrap p=0.0060).
>
> **Limits.** cross-country association; no demonstrated within-Greece dynamic; no causal claim; current conditions not ruled out; supported only as the CURRENT uninterrupted run against a fixed 2008 base; alternative constructions point the same way but do not meet the criteria.

> Housing-cost deterioration since 2010 predicts hardship (bootstrap p=0.0460) and retains a cross-country association after its current-level counterpart is controlled

> (conditional bootstrap p=0.0015).
>
> **Limits.** cross-country association; no demonstrated within-Greece dynamic; no causal claim; current conditions not ruled out; BORDERLINE: bootstrap p=0.0460, 91 of 1,999 exceedances.

The third is borderline, and is labelled as such: 91 of 1,999 bootstrap replications exceeded the observed statistic. It is reported at the strength the test gives it and no more. The second is supported only under one specific construction — the current uninterrupted run measured against a fixed 2008 base. Alternative constructions of the same idea point the same way but do not meet the pre-registered criteria, and they are not counted as corroboration.

T2 **Each accumulated measure tested with its own current-level counterpart in the same model.**

ACCUMULATED MEASURE	CONTROLLING FOR	COEFFICIENT	BOOTSTRAP P	VERDICT	CEILING APPLIED
Accumulated unemployment	Long-term unemployment	+0.2936	0.0025	**Supported**	—
Cumulative wage shortfall	Real wages	+0.1094	0.0060	**Capped by ceiling**	yes
Cumulative GDP shortfall	Share below own GDP peak	+0.1057	0.1155	**Inconclusive**	—
Cumulative threshold shortfall	Real poverty threshold	+0.1848	0.0755	**Inconclusive**	—
Accumulated affordability pressure	Wage-adjusted affordability	+0.0124	—	**Inconclusive**	—
Compounded inflation since 2008	Inflation	-0.1667	—	**Inconclusive**	—
Housing deterioration since 2010	Housing-cost overburden	+0.2423	0.0015	**Supported**	—

The ceiling column marks where Stage 5 was barred from creating support that Stage 4 had not established.

The pattern is not uniform across constructs, which is itself informative.

> For wage-adjusted affordability the pattern reverses: the current measure survives conditioning (bootstrap p=0.0035) while the accumulated one remains inconclusive.
>
> **Limits.** the accumulated measure is inconclusive, NOT unsupported.

For wage-adjusted affordability the relationship runs the other way: the present-day measure survives conditioning while the accumulated one does not resolve. If accumulation were a general property of this outcome, this reversal should not appear. It suggests the accumulated measures are picking up something specific to labour markets, wages and housing rather than a universal history effect.

The limit that must not be crossed

Everything above is a statement about differences *between countries*. It is tempting, and wrong, to convert it into a statement about change *within* Greece over time.

FIGURE 13 **The supported historical associations are predominantly between countries; the within-country estimates provide no supporting dynamic evidence**

PRE-REGISTERED CONDITIONAL ROBUSTNESS Is the effect between countries or within them?

Read with this. This is THE central limitation: no within-country estimate is significant in the adverse direction and no first-difference test supports one. The first view is STANDARDISED, and each component by its own spread: between-country and within-country variation differ in size by factors of 0.8 to 5.7 here, so one shared SD would distort the comparison it is meant to fix. The second view shows first differences in each measure's own units, which is why those bars carry no shared scale and are read one row at a time. This does NOT establish that no within-country relationship exists: the within estimates are too imprecise to establish or rule out such a relationship. They are recorded as inconclusive, not absent.

Between vs within

Accumulated measure	Between (SD)	Within (SD)	Between p	Within p	First difference (raw)	First difference p
Accumulated unemployment	0.5924	0.0296	0.0000	0.6867	0.0461	0.7324
Cumulative wage shortfall	0.6584	-0.0544	0.0130	0.3147	-0.0596	0.0284
Years wages below 2008	0.7271	0.0700	0.0025	0.0823	0.0721	0.4952
Cumulative GDP shortfall	0.3890	0.0674	0.0289	0.0600	0.0273	0.1442
Cumulative threshold shortfall	0.8271	0.0123	0.0000	0.5599	-0.0148	0.3917
Accumulated affordability pressure	-0.3953	-0.0764	0.3639	0.2825	-0.0298	0.0000
Compounded inflation since 2008	-0.1718	-0.1228	0.4947	0.3738	-0.0314	0.5766
Housing deterioration since 2010	0.9282	-0.0658	0.0000	0.2343	-0.0090	0.8679

Year-on-year changes

Accumulated measure	First difference	p	n
Accumulated unemployment	0.0461	0.7324	243.0000
Cumulative wage shortfall	-0.0596	0.0284	243.0000
Years wages below 2008	0.0721	0.4952	243.0000
Cumulative GDP shortfall	0.0273	0.1442	243.0000
Cumulative threshold shortfall	-0.0148	0.3917	234.0000
Accumulated affordability pressure	-0.0298	0.0000	243.0000
Compounded inflation since 2008	-0.0314	0.5766	243.0000
Housing deterioration since 2010	-0.0090	0.8679	242.0000

> No accumulated measure permits dynamic wording. Across P5, E4 and E7 no within-country estimate is significant in the adverse direction and no first-difference test supports one.
>
> **Limits.** three RELATED checks on one panel, not independent replications; the dynamic tests were not multiplicity-corrected.

> **This is the report's most easily overstated result.** No within-country estimate reaches significance in the adverse direction, and no first-difference test supports a dynamic reading. But imprecise estimates are not proof of absence: the within-country tests have considerably less power than the between-country ones, because they discard all the cross-sectional variation. The correct statement is that this design does not support dynamic wording — not that the dynamic effect has been shown to be absent. Three separate drafts of this report crossed that line before it was caught.

One construct could not be tested at all, and is reported rather than quietly dropped.

> Accumulated material resources (C1) could not be tested at all: the source series begins in 2015 and no 2008 baseline exists. The baseline was not moved to make it testable.
>
> **Limits.** infeasible, not null.

Moving the baseline to 2015 would have made it testable and would also have made it a different measure, one that could not register the crisis it was built to capture. The baseline was left where it was.

One further result is reported here without being counted. The accumulated wage-shortfall measure passed every conditional test it faced — but the present-day measure it was paired with had not been supported in the previous stage, and a rule fixed in advance prevents this stage from promoting anything its predecessor did not establish. Reported, and not a finding.

> The accumulated wage-shortfall coefficient cleared every conditional gate but its prior stage did not support it, so the pre-registered ceiling caps it. It is reported and is not a finding.
>
> **Limits.** capped by the E7 ceiling: E7 may only qualify or withdraw.

The rule costs something here: on the face of the numbers this measure looks as convincing as the three that are counted. The reason for keeping the rule anyway is that conditional tests run after seeing which measures succeeded are selected tests, and a rule that can be set aside when its result is inconvenient is not a rule.

– Accumulation, conditioning, the Mundlak decomposition, and the E7 ceiling

HOW ACCUMULATION IS COMPUTED

Each accumulated series is a cumulative sum of annual excess over a fixed baseline. The property that matters is that no accumulated value at year *t* may use information from any year after *t* — otherwise a measure of history would quietly contain the future. This is checked by rebuilding each series from truncated inputs: if any value changes when later years are withheld, the construction leaks.

One construction error was caught this way. The wage-adjusted excess measure was written to subtract 100 from an index, but the underlying source held absolute euro values near 32,000 rather than an EU27=100 index, so the excess was zero in every country and year. The series was silently degenerate and every downstream test on it was meaningless. It is now indexed explicitly, with an assertion that fails loudly if the source scale changes again.

WHAT CONDITIONING MEANS HERE

The conditional test places the accumulated measure and its current-level counterpart in the same specification, alongside AROP and year effects, and asks whether the accumulated coefficient survives. Because the two are correlated by construction, this test is conservative: shared variance is attributed to neither, so an accumulated measure can only clear it by carrying information the current level does not.

Minimum detectable effects in this stage are computed pair-specifically and conditionally — the MDE for accumulated unemployment given current unemployment, not in isolation. An unconditional MDE would understate how demanding this test is and would make the inconclusive results look like stronger nulls than they are.

THE BETWEEN/WITHIN DECOMPOSITION

The dynamic question is addressed with a Mundlak specification, which enters both each country's mean of a predictor and its deviation from that mean. The coefficient on the country mean is the between-country association; the coefficient on the deviation is the within-country one. Reporting only the pooled estimate would blend the two and allow a between-country result to be read as a within-country process.

The within estimates in this report are imprecise. That is a property of the data — roughly a decade of annual observations per country, with slow-moving series — and not a finding. The figure shows both estimates with their intervals so that the imprecision is visible.

THE E7 CEILING

This stage may only qualify or withdraw support that Stage 4 established. It may never create support. The reason is that these conditional tests are run after seeing which measures succeeded, which makes them selected tests: a measure that reaches this stage has already passed a filter, so letting the stage promote it would turn a selected result into a finding.

The limit binds in exactly one case, the accumulated wage-shortfall measure, which cleared every conditional test while its present-day counterpart had not been supported. It is reported in Stage 5 and it is not counted.

Where this leaves us. Accumulated exposure on labour, wages and housing carries cross-country information beyond present conditions. No dynamic claim about Greece over time is supported. The next stage asks how much of all this survives a change in one modelling decision — and the answer is uncomfortable.

STAGE 6 How much depends on the model

Would a defensible alternative specification have produced a different answer?

Most empirical reports include a robustness section demonstrating that the findings hold up. This one includes a robustness section demonstrating that one of them does not, and that fact is a result rather than an embarrassment.

Recall from Stage 3 that four deprivation items absorbed most of Greece's excess residual, and that they share their instrument with the outcome. Whether to admit such a predictor is a real methodological choice, and the data cannot settle it.

The case for admitting it: severe material deprivation is a substantively meaningful measure of hardship, it is officially published, and excluding variables because they are measured well is not a principle anyone actually holds. The case against: when predictor and outcome come from the same respondents answering adjacent questions in one sitting, part of any relationship between them reflects shared measurement rather than shared substance, and a model that leans on it will attribute to explanation what is really instrument.

Both positions are defensible. So we ran both.

FIGURE 14 **Removing the same-instrument deprivation measure moves Greece from under-predicted to over-predicted**

POST-SELECTION ROBUSTNESS Does the answer depend on which model is chosen?

Read with this. NEITHER specification is definitive. They may not be merged or averaged, and selection may not be made on residual size.

Country	Reference specification	Deprivation-free companion	Change
Luxembourg	8.80	15.48	6.68
Cyprus	7.05	10.75	3.70
Greece	6.93	-9.39	-16.32
Latvia	6.32	4.65	-1.66
Hungary	4.86	7.42	2.56
Bulgaria	4.06	14.65	10.60
Czechia	3.53	-3.63	-7.16
Slovakia	3.28	-0.68	-3.96
Belgium	3.13	5.26	2.13
Ireland	3.05	6.14	3.09
Croatia	2.62	-0.38	-2.99
France	1.85	3.88	2.02
Austria	1.36	0.95	-0.41
Portugal	1.30	0.62	-0.67
Malta	1.28	0.36	-0.91
Poland	0.56	-4.32	-4.88
Denmark	0.16	-3.65	-3.81
Sweden	-1.00	-4.55	-3.55
Slovenia	-2.66	-6.33	-3.67
Netherlands	-3.47	-4.87	-1.39
Lithuania	-4.17	-1.81	2.36
Germany	-4.31	-6.00	-1.69
Estonia	-4.49	-10.12	-5.63
Finland	-5.07	-6.80	-1.72
Italy	-6.35	-5.18	1.17
Romania	-10.32	9.76	20.08
Spain	-10.63	-10.06	0.57

> Greece's residual is +6.93 (rank 3/27) in the reference specification and -9.39 (rank 25/27) when the same-instrument deprivation predictor is removed, on identical rows. Neither specification is definitive.
>
> **Limits.** NEITHER specification is definitive; the two may not be merged or averaged; selection between them may not be made on residual size; conclusions about absorption depend materially on whether same-instrument deprivation is admitted.

T3 The same rows, one predictor, two opposite conclusions.

SPECIFICATION	SAME-INSTRUMENT DEPRIVATION	GREEK RESIDUAL	RANK	READING
Reference specification	included	+6.93	3 of 27	Greece markedly worse than pre
Companion specification	removed	−9.39	25 of 27	Greece markedly better than pre

Neither specification is definitive. The two may not be averaged, and the choice between them may not be made on residual size.

Greece moves from third-highest unexplained hardship in Europe to twenty-fifth — from a marked positive anomaly to a marked negative one — on identical rows, with one predictor added or removed. This is not a small sensitivity. It is a reversal.

> **Neither specification is definitive, and they may not be reconciled.** The two results may not be averaged, and the choice between them may not be made by looking at which produces a more plausible residual. Selecting a specification on the size of its residual is exactly the practice that makes robustness checks meaningless. What the report can say is that conclusions about how much of Greece's anomaly is absorbed depend materially on a modelling decision that the data cannot adjudicate, and any reader should treat absorption results with that in mind.

This is why Stage 3's absorption result is framed as a diagnostic rather than an explanation, and why the conclusion in Stage 8 does not rest on it. A finding that reverses under a defensible alternative is not a finding to build on.

Two analyses that were planned and failed

The report would look stronger without this section, which is the reason it is here. Both of the analyses below were specified in advance, both were meant to carry weight, and neither worked.

The first was a synthetic control, and it was intended as the centrepiece: build a weighted combination of other countries that tracks pre-crisis Greece, then read the divergence after 2008 as the crisis effect. The construction failed. The donor weights collapsed onto two countries, so the synthetic Greece was a blend of Hungary and Bulgaria rather than a credible counterfactual, and the design missed four of its six pre-registered conditions.

> The synthetic-control comparative design failed four of six pre-registered gates and is not a usable comparison. Its divergence figure is excluded from every document in this project.

> **Limits.** donor weights collapse to Hungary 0.55 and Bulgaria 0.45; the divergence figure is NON-REPORTABLE.

Its divergence chart would be the most arresting image in this report. It is not shown anywhere, and that is deliberate: a striking picture built on a counterfactual this thin would persuade readers of something the analysis cannot support. Being compelling is not the same as being right, and the first is more dangerous when the second is absent.

The second was a measure of how many domains of disadvantage a country shows at once. Adding it to the reference model made Greece's residual worse rather than better, and its sign flipped once other measures were controlled. That reversal is left uninterpreted here. An unexplained sign flip in a specification that has already failed is exactly the kind of loose end that invites a story to be built around it, and the story would be untestable.

> Multi-domain breadth failed the incremental criterion: adding it to the reference model worsened Greece's residual from 6.93 to 10.39 and reversed its sign conditionally. The reversal is left uninterpreted.
>
> **Limits.** a result about the specification, not about power; the sign reversal is deliberately left uninterpreted.

– The two specifications, and how their residuals are compared

THE REFERENCE SPECIFICATION

The reference specification regresses reported hardship on income poverty, the supported present-day measures, severe material deprivation and year fixed effects, and extracts each country's residual. It was fixed before the final results were produced. The companion specification is identical except that the same-instrument deprivation predictor is removed.

Both are run on **identical rows**. A common way to manufacture an apparent robustness failure, or to conceal one, is to let the estimation sample shift when a variable with different coverage enters or leaves. The row set is fixed to the intersection before either model runs, and this is asserted in code rather than checked by eye.

HOW THE RESIDUALS ARE EXTRACTED

A country residual is the average, across that country's years, of the difference between its observed hardship rate and the rate the model predicts. A positive residual means a country reports more hardship than its characteristics predict; a negative one means less. Ranks are taken across all twenty-seven countries within each specification separately, so a rank change reflects Greece's movement relative to the same comparison set rather than a change in who is being compared.

Residuals are averaged over years rather than taken from a single year, because a single-year residual is sensitive to which year is chosen and would make the comparison in this stage look more or less dramatic depending on an arbitrary decision. Year fixed effects are already in both models, so the averaging is over deviations that have had Europe-wide shocks removed.

WHY RESIDUAL SIZE MAY NOT SELECT A SPECIFICATION

Both models produce a Greek residual, and it would be easy, and entirely circular, to prefer whichever one better matched a prior expectation about Greece. Choosing between them on the size of the residual is therefore ruled out. That is why this report presents both and settles on neither.

WHY THE FAILED DESIGN'S CHART IS ABSENT

A synthetic control whose donor pool collapses onto two countries does not describe a counterfactual Greece, so the gap between the real and synthetic series measures the failure of the construction rather than the effect of the crisis. Plotting it would give a precise-looking magnitude to a quantity that has no defensible interpretation.

Where this leaves us. The empirical sequence is complete. Before drawing conclusions, one alternative explanation needs addressing directly — that Greeks simply report everything more negatively — along with the contextual factors this project did not test and must not pretend it did.

STAGE 7 What this is not

Is this a reporting artefact, and what else might matter that this project did not test?

The most economical explanation of everything so far is that Greeks answer survey questions more negatively than other Europeans. If that were true, the gap in Stage 1 would be a property of the respondents rather than of their circumstances, and much of what follows would be measuring national temperament.

Stage 3 partly addressed this: reported difficulty tracks concrete affordability failure. But that evidence came entirely from within one survey instrument, so it cannot settle the question by itself. A better test compares Greek responses across *different domains*. A general negative reporting tendency should depress everything roughly equally. A domain-specific pattern — extreme on financial questions, less extreme elsewhere — points toward circumstances rather than temperament.

INDICATOR	UNIT	GREECE	EU MEDIAN	GREECE'S POSITION, WORST FIR	COUNTRIES
Reported hardship	%	66.7	17.1	1	27
Financial expectations	index	-43.2	-1.8	1	27
Life satisfaction	index	6.7	7.3	2	28

2024. A percentage, a net balance and a 0-10 rating do not share a scale, so level and position are given rather than plotted together. The distributions themselves are charted in the statistical appendix.

The level and the position answer the question together. On the two money questions Greece is not merely last: it sits clear of the cluster, at 66.7% reporting difficulty against a European median of 17.1%, and at −43.2 on expectations against a median of −1.8. On life satisfaction Greece is inside the pack, at 6.7 against a median of 7.3, and second from the worst end.

> Greece is consistently among Europe's worst countries on life satisfaction and consistently worst on financial hardship and expectations. The greater extremity of the financial indicators suggests domain specificity, but does not rule out a broader negative reporting tendency.
>
> **Limits.** Greece is second-WORST on life satisfaction by 2024, not middling; a broader negative reporting tendency is NOT ruled out; Greek life satisfaction ROSE over the observed

> period, 6.2 to 6.9; only its rank worsened, because other countries improved faster; the series begins in 2013, at the crisis trough, with no pre-crisis baseline.

So the pattern is domain-specific and the difference is one of degree, not of kind. That is not the profile of a country whose financial reports are extreme while its general outlook is ordinary, and an earlier draft of this report described it that way in error.

Two things must be held apart here, and conflating them is easy. Greek life satisfaction *rose* over the observed period, from 6.2 to 6.9. Greece's *rank* nonetheless worsened, because other countries improved faster. A worsening rank on a rising series is not falling satisfaction, and reading it as such would invert the finding.

DESCRIPTIVE CORROBORATION

FINANCIAL EXPECTATIONS AND LIFE SATISFACTION

The cross-domain comparison is descriptive corroboration for the domain-specificity reading, not a test of it. Greece's financial indicators sit at the extreme of the European distribution while its general-wellbeing indicator sits close to it, and that difference in degree is what the figure above shows.

It does not rule out a broader negative reporting tendency operating alongside genuine financial difficulty. Both could be true at once, and this design cannot separate them.

What may be concluded. The financial indicators are more extreme than the general-wellbeing one, which suggests domain specificity. A broader negative reporting tendency is NOT ruled out.

Limitation. Describing Greece as ordinary or middling on life satisfaction: it is second-worst in the EU by 2024. And reading a worsening RANK as falling satisfaction, when the Greek level rose over the period.

Source. reporting_style_cross_indicator.csv, this project Greece ranks 1st of 27 on hardship and financial expectations, 2nd-6th on life satisfaction.

A separately labelled descriptive extension

The Eurostat series used everywhere else in this report begins in 2013, at the crisis trough, so it cannot say whether Greece was already unusual before 2008. The European Social Survey can. What follows is kept deliberately apart from everything above: a different instrument, shown on its own, never joined to the EU-SILC series and never modelled.

It also carries a weaker warrant than the rest of the report, and the difference matters. The respondent-level files are behind a login. What is used here comes from the ESS portal's public analysis view, which displays weighted response distributions by country; country means were reconstructed by multiplying each displayed percentage by its score and summing. The portal rounds those percentages, so the means are approximate at that precision. There are no standard errors and no confidence intervals, and nothing inferential is built on them.

Six observations across two decades, with a ten-year hole in the middle, do not make a line. They make a short table, which can carry the gap as a stated fact rather than as a break in a chart.

Greek life satisfaction across six ESS rounds, against the median of the same twelve countries in every round. Means are approximate reconstructions from published weighted percentages: no interval can be attached to them and nothing is tested on them.

PERIOD	FIELDWORK	GREECE	MEDIAN OF THE 12	GAP	RANK OF 12
pre-crisis	2002/03	6.35	7.21	-0.86	3
pre-crisis	2004/05	6.46	7.25	-0.80	4
crisis onset	2008/09	5.96	7.09	-1.13	3
early crisis	2010/11	5.64	7.00	-1.36	1 (worst)
later period	2020-22	6.35	7.48	-1.12	1 (worst)
later period	2023/24	6.42	7.39	-0.97	1 (worst)

No Greek round falls between 2010/11 and 2020-22, so the decade covering the depth of the adjustment and the recovery is unobserved.

Three things happen in sequence, and separating them is the point. Greece was *already* about 0.8 points below the median of these twelve countries before the crisis, at third or fourth worst. Its level then fell to 5.64 by 2010/11. And by 2023/24 it had recovered to 6.42, close to where it started.

Its comparative position did not recover with it. The gap to the median is wider now than before the crisis, and Greece has been the worst of the twelve since 2010/11 — the two countries that were below it before the crisis have since passed it. Part of that is the median rising; part is being overtaken.

> **What this does to the reporting-style question.** A low Greek baseline pre-dates the crisis, so the position cannot be attributed to the crisis alone, and a longstanding low-wellbeing pattern remains plausible. Generic pessimism or reporting culture cannot be dismissed on this evidence — and cannot be established by it either. It sharpens the caveat already attached to the claim above rather than replacing it.

DESCRIPTIVE CORROBORATION

PRE-CRISIS WELLBEING BASELINE (ESS)

Across six ESS rounds, holding the same twelve countries fixed, Greece sat roughly 0.8 points below the median before the crisis, fell to 5.64 by 2010/11, and recovered its level to approximately its pre-crisis value by 2023/24 while its comparative position did not recover.

The all-country ranks in the source data are deliberately not used for comparison here. The full ESS country set varies from twenty-two to thirty between rounds, so an all-country rank moves when the country set moves, and a change in it would be partly composition rather than substance.

What may be concluded. On a balanced set of 12 countries observed in every Greek ESS round, Greece already sat about 0.8 points below the median before the crisis, fell to 5.64 by 2010/11, and recovered its LEVEL to roughly the pre-crisis value by 2023/24. Its comparative position did not recover: the gap to the balanced median is wider than before the crisis, and Greece has been worst of the 12 since 2010/11. A longstanding low-wellbeing pattern therefore remains plausible and generic pessimism or reporting culture CANNOT be dismissed.

Limitation. Splicing ESS to the Eurostat EU-SILC series, or reading the two as one trajectory. Attaching any confidence interval, standard error or significance test to these means, which are approximate reconstructions from displayed weighted percentages. Comparing ALL-COUNTRY ranks across rounds, since the country set varies from 22 to 30. Treating the decade after 2010/11 as observed.

Source. European Social Survey Data Portal, public Analysis tab for stflife (life satisfaction, 0-10), rounds 1, 2, 4, 5, 10 and 11. Weighted distributions with the post-stratification weight; country means reconstructed by the authors. source

Health: a second post-freeze extension

Health is the other obvious candidate a reader will ask about, and it was examined after the claim set was frozen. Like the survey extension above, it is kept apart from everything in Stages 1 to 6, and for the same reason: it could not have changed a headline claim whatever it returned.

The finding worth carrying forward is descriptive and it is stark. On unmet medical care — the share who needed care and did not get it because of cost, waiting time or distance — Greece was second-worst in the Union in 2016, 2019 and 2022, and worst of twenty-seven in 2024, at 12.1% against a median member state of 1.9%. That is a health-system fact of the same order as anything in Stage 1, and it belongs in any account of what hardship means in practice for a Greek household.

What health does *not* do is explain the gap. Four measures were tested against the same baseline used elsewhere — unmet care, self-rated health, long-standing illness, and activity limitation — and none survives. Three of the four carry the wrong sign: across countries, worse reported health goes with *less* reported hardship, which is not a finding but a warning. Separating the comparison shows why. All three health-status measures reverse sign between countries and within them, so the pooled coefficient describes neither comparison; within a country, years of worse reported health are years of more reported hardship, the expected direction. National composition and reporting conventions dominate the pooled association. Unmet care is the exception that makes the point: it measures access rather than health, it is positive in both comparisons, and it is also the one measure whose leave-one-country-out refits change sign.

So health enters this report as context and as a research direction, not as a predictor. The natural next step is not more EU-SILC: it is age-standardised rates and independent sources — out-of-pocket and catastrophic health spending, medicine access, avoidable mortality — which could test whether the survey signal matches administrative evidence instead of restating the same instrument. The full analysis, its eight artifacts and three figures are in the statistical appendix and in `docs/health_preliminary_analysis.md`.

What this project did not test

Several factors are prominent in accounts of the Greek crisis and are absent from the analysis above. Omitting them silently would be misleading; discussing them as though they were findings would be worse. They are recorded here in a separate register, with an explicitly different vocabulary, and each entry states what it permits and what it forbids.

The boundary is not that none of it was examined — migration was tested diagnostically on this panel, and the cross-domain comparison and the ESS extension are both descriptive analyses carried out here. The boundary is that **none of it may support a headline claim**. These entries carry their own status vocabulary, deliberately disjoint from the evidence statuses used in Stages 1 to 6, so that a contextual statement can never be promoted into a finding.

T5 **Material this report discusses without establishing.**

TOPIC	WHAT IT IS	CAN IT SUPPORT A FINDING?
Financial expectations and life satisfaction	descriptive corroboration	no
Institutional trust	contextual evidence	no
Crisis and adjustment policies	literature-grounded context	no
Anchored poverty, corroborated by independent microdata	descriptive corroboration	no
Migration	contextual consequence	no
Tax burden and unequal treatment	future hypothesis	no
Policy implications	author interpretation	no
Pre-crisis wellbeing baseline (ESS)	descriptive corroboration	no
Independent descriptive corroboration (Greece in Figures)	descriptive corroboration	no
Wealth concentration since 2019	contextual evidence	no

Some of these were examined here — migration was tested on this panel, and the cross-domain and ESS comparisons are descriptive analyses done for this report. None of them can carry a finding, and their statuses come from a deliberately separate vocabulary from the evidence statuses above so the two cannot be confused.

CONTEXTUAL EVIDENCE

INSTITUTIONAL TRUST

Institutional trust is low in Greece by OECD measurement, and there is a plausible route by which it could matter: households that do not expect effective support may experience the same material circumstances as more threatening, and report accordingly. The pattern within Greece is uneven, which matters for that reading: the institutions a household would actually turn to for support are not the ones Greeks trust most.

This project ran no test on trust. What the OECD's own country note for Greece reports — charted in the statistical appendix under contextual evidence — is more interesting than a single comparison: Greek trust is not uniformly low. Greeks report more trust in the police and the courts than in any elected or political institution, and political parties sit at the bottom at 17%. Central government, at 32% against an OECD average of 39%, sits in the middle of a wide internal range rather than at the bottom of it.

What may be concluded. May affect how households interpret insecurity; available evidence does not identify an independent effect.

Limitation. Presenting trust as an explanation of the residual, or implying it was tested and found to matter.

Source. OECD (2024), OECD Survey on Drivers of Trust in Public Institutions - 2024 Results, Country Notes: Greece. OECD Publishing, Paris. Fieldwork October-November 2023, 30 OECD countries. 32% of Greek respondents report high or moderately high trust in central government against an OECD average of 39%. source

LITERATURE-GROUNDED CONTEXT

CRISIS AND ADJUSTMENT POLICIES

The adjustment programmes of 2010 onward reshaped Greek incomes, employment protection, pensions and public services simultaneously and over a compressed period. They are the historical setting for every accumulated measure in Stage 5.

This project estimates no policy effect. The accumulated measures record what a country absorbed; they do not attribute any of it to a specific programme, measure or decision, and the design contains nothing that would license such an attribution.

What may be concluded. Help explain the historical setting; this project does not estimate their causal contribution.

Limitation. Attributing any share of the accumulated exposure to a specific programme or measure.

Source. Andriopoulou, E., Kanavitsa, E. & Tsakloglou, P. (2020), Decomposing Poverty in Hard Times: Greece 2007-2016. LSE GreeSE Paper No. 149. EU-SILC microdata, ELSTAT waves 2008-2017. Anchored FGT0 on a 2007 base reaches 48% at the 2013 peak. source

CONTEXTUAL CONSEQUENCE

MIGRATION

Emigration of working-age Greeks during the crisis is well documented, and it interacts with the labour-market measures in Stage 4 in both directions: it is plausibly a consequence of prolonged labour-market damage, and plausibly a contributor to the composition of who remains.

Migration was tested on this panel as an aggregate predictor in Stage 4 and found nothing ($p=0.4006$). That result speaks to aggregate prediction only. It does not establish that migration is irrelevant, and it does not speak to either causal direction.

What may be concluded. A plausible consequence of prolonged labour-market damage and a possible contributor to it. It is UNSUPPORTED as an independent aggregate predictor here.

Limitation. Reading it as a driver, or as ruled out. E3 tested it on this panel and found nothing (p=0.4006), which speaks to aggregate prediction and not to either causal direction.

Source. Lazaretou, S. (2016), The Greek brain drain: the new pattern of Greek emigration during the recent crisis. Economic Bulletin, Bank of Greece, issue 43, pp. 31-53. 427,000 residents aged 15-64 left permanently 2008-2013; ~223,000 of them aged 25-39. The E3 null (p=0.4006) is this project's own evidence. source

FIGURE 15 **The crisis also became an exit route — and since 2023 that has reversed**

CONTEXTUAL CONSEQUENCE Did the crisis also become an exit route, and has that reversed?

Read with this. CONTEXTUAL CONSEQUENCE AND POSSIBLE CONTRIBUTOR, not an independently supported predictor. Net migration was tested directly at the diagnostic stage and returned nothing (p = 0.4006), which speaks to aggregate prediction and to neither causal direction. Net outflow peaked at 44,502 in 2012. Across 2008-2024 the cumulative net loss is 290,281 people, 2.69% of average population and the fifth highest of the 25 countries with sufficient coverage. By 2023 the flow had turned: more Greek nationals returned than left, and in 2024 the net return was 19,852. The accumulated loss is now beginning to reverse; it has not been undone.

Departures and returns

Series	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Departures	19088	19799	28301	61268	70696	67057	59525	60516	61368	60816	57495	53415	41111	41505	40151	36931	32141
Returns	25070	22790	25078	27043	26194	26644	29503	30460	30747	31743	32199	34074	20864	28392	32017	46091	51993

Net flow

Series	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Net outflow of nationals	-5982	-2991	3223	34225	44502	40413	30022	30056	30621	29073	25296	19341	20247	13113	8134	-9160	-19852

EU comparison, cumulative

Series	Cumulative net departures	% of average population	Years covered
Lithuania	243338.00	8.27	17.00
Croatia	243289.00	5.93	17.00
Romania	849659.00	4.30	17.00
Luxembourg	17038.00	2.96	17.00
Greece	290281.00	2.69	17.00
Poland	1014893.00	2.69	16.00
Slovenia	46557.00	2.25	17.00
France	1380603.00	2.08	17.00
Ireland	90151.00	1.87	17.00
Belgium	198811.00	1.75	15.00
Estonia	21929.00	1.65	17.00
Italy	799607.00	1.34	17.00
Sweden	127581.00	1.29	17.00
Austria	98950.00	1.14	17.00
Germany	882086.00	1.07	17.00
Netherlands	173972.00	1.02	17.00
Portugal	83133.00	0.80	16.00
Spain	214223.00	0.46	17.00
Czechia	48258.00	0.46	17.00
Finland	23245.00	0.43	17.00
Slovakia	-936.00	-0.02	17.00
Hungary	-30075.00	-0.31	17.00
Malta	-1979.00	-0.43	17.00
Denmark	-68217.00	-1.20	17.00
Cyprus	-11840.00	-1.35	16.00

FUTURE HYPOTHESIS

TAX BURDEN AND UNEQUAL TREATMENT

Published incidence research establishes that the Greek indirect tax system became markedly more regressive across the crisis, shifting burden toward lower-income households. If a household's disposable position worsened through taxation in ways that income-based poverty measures capture poorly, that would be one route to a gap of the kind this report documents.

This is a hypothesis for future work and nothing more. The cited literature concerns tax incidence, not subjective hardship; this project tested nothing on tax; and no quantitative link between tax burden and the hardship gap is asserted or implied.

What may be concluded. Published incidence evidence establishes that Greece's indirect tax system became markedly more regressive over the crisis. Whether that channel contributes to the hardship gap is a plausible hypothesis and is NOT tested here.

Limitation. Any quantitative statement linking tax burden to the hardship gap. The literature is about incidence, not about this outcome, and this project tested nothing on tax.

Source. Kaplanoglou, G. (2015), Who Pays Indirect Taxes in Greece? From EU Entry to the Fiscal Crisis. Public Finance Review 43(4), 529-556. Microsimulation on Household Expenditure Survey data, 1988-2011. The 2011 indirect tax system is the most regressive of the period, with the sharpest effects on families with children and the unemployed. source

CONTEXTUAL EVIDENCE

WEALTH CONCENTRATION SINCE 2019

ECB distributional wealth accounts show that of the €224.8bn increase in Greek household net wealth between 2019 Q2 and 2026 Q1 (+32.2%), about 72.6% accrued to the wealthiest fifth of households, while the bottom half accounted for about 9.2% and its own wealth share slipped from roughly 9.5% to 9.4%. This is one plausible mechanism for how aggregate recovery and persistent reported hardship coexist.

This project ran no test linking wealth concentration to reported hardship, and the wealth accounts sit partly outside the 2015-2024 window used elsewhere in this report. A rising net-wealth total also does not mean rising disposable income: these changes are driven mostly by property and financial-asset valuations, not household cash flow, and the ECB itself labels the underlying accounts experimental.

What may be concluded. Independent ECB distributional wealth accounts show that of the €224.8bn increase in Greek household net wealth between mid-2019 and early 2026 (+32.2%), about 72.6% accrued to the wealthiest fifth of households while the bottom half accounted for about 9.2%; the bottom half's own share of total wealth slipped slightly, from roughly 9.5% to 9.4%. This offers one plausible mechanism for how aggregate recovery and persistent reported hardship can coexist: a rising net-wealth total does not evenly reach the households reporting the most difficulty.

Limitation. Treating this as tested against this project's reported-hardship findings, or as an established explanation for the 52.6-point gap: it sits partly outside the 2015-2024 window and was never checked against any model here. Reading a rising net-wealth figure as equivalent to rising disposable income or an improved ability to pay bills: these changes are driven mostly by property and financial-asset valuations, not household cash flow. Treating the ECB's distributional wealth accounts as final: the ECB itself labels them experimental, combining household survey data with quarterly national accounts and extrapolating between survey waves.

Source. European Central Bank, Distributional Wealth Accounts (DWA), Greece, household sector (S14), 2019 Q2–2026 Q1. Net wealth rose from €698.4bn (2019 Q2) to €923.2bn (2026 Q1), +32.2%. Of the €224.8bn increase, the top decile alone gained €133.4bn; the top two deciles combined gained 72.6% of the total increase, the 6th-7th deciles 9.7%, and the bottom half 9.2%. A companion Eurostat check over the same window (namq_10_gdp)

puts Greek real GDP growth at +11.7%, ranking 20th of 36 countries with both observations.
source

How the cross-domain comparison works, and what it cannot settle

THE COMPARISON

Three Eurostat indicators are compared for Greece against the other EU member states: reported financial hardship, financial expectations, and general life satisfaction. The first two are financial; the third is not. If a country scores at the extreme of the distribution on all three by a similar margin, that is consistent with a general reporting tendency. If it is extreme on the financial pair and less so on the general one, that is consistent with domain specificity.

Greece is worst in Europe on the financial pair and second-worst on life satisfaction by 2024. The difference is a difference of degree, and the report describes it that way.

RANK AND LEVEL ARE DIFFERENT QUANTITIES

The figure shows rank trajectories because the question is comparative, and rank is treacherous on its own for the reason already noted above (a score can improve while its rank still worsens). The figure carries the underlying values alongside the ranks for that reason, and the axis is labelled so that the worst position is unambiguous.

WHAT THIS CANNOT SETTLE

Separating genuine financial difficulty from genuine difficulty *plus* a general negative tendency requires either a pre-crisis baseline, which this series does not have, or an external anchor on Greek response style, which this project does not have. The claim is worded to leave both possibilities open.

HOW THE ESS FIGURES WERE RECONSTRUCTED

The respondent-level ESS files require an account. The portal's public analysis view does not, and it publishes, for each country and round, the weighted percentage of respondents choosing each point on the 0–10 life satisfaction scale. A country mean is recovered from that by multiplying each percentage by its score and summing.

The portal rounds the percentages it displays, so the recovered means are approximate at that precision. More importantly, a distribution published as percentages carries no information about sampling variability: there is no way to attach a standard error or an interval to a mean recovered this way, and no test is run on them anywhere in this report.

The twelve-country balanced set exists because the full ESS country set ranges from twenty-two to thirty across the six Greek rounds. A rank computed against a changing set moves when the set moves, so an all-country rank trajectory would confound Greece's position with who happened to participate. Holding the same twelve countries fixed removes that, at the cost of comparing Greece against a smaller and richer group than the EU average used elsewhere — which is one more reason the two series are never placed on the same axis.

WHY TRUST WAS NOT TESTED

Institutional trust is measured by the OECD on a different instrument, a different sample and a different periodicity from the Eurostat panel used throughout this report. There is no defensible way to merge them at country-year level for the period analysed, and constructing one would manufacture the comparability it needs. Trust therefore appears as registered context, and the figure reports only the two values read directly from the primary OECD release.

Where this leaves us. The gap is not adequately explained as a reporting artefact, though a general negative tendency cannot be excluded. Several plausible contributing factors were not tested and are recorded as untested. What remains is to state the conclusion at the strength the evidence actually supports.

STAGE 8 What the evidence supports

Stated no more strongly than the tests allow.

T4 Everything the analysis established, and everything it did not.

STATUS	WHAT IT CONCERNS	WHERE AND HOW
Supported	Material resources, long-term unemployment and wage-adjusted affordability predict hardship beyond income poverty	Stage 4, bootstrap 0.0005–0.0085
Supported	Accumulated unemployment, wage non-recovery and housing deterioration add cross-country information beyond their current-le	Stage 5, conditional bootstrap 0.0015–0.0060
Descriptive corroboration	Reported hardship co-moves with concrete affordability failure within countries — all items from the same EU-SILC interview, so th	Stage 3, within-country r = 0.63–0.80
Inconclusive	Six of nine current-level constructs, and the accumulated wage-adjusted measure	Stage 4 and Stage 5, power-limited
Unsupported	Annual food and housing inflation at 0.70 SD; annual headline inflation at its detectable specific	Stage 4, magnitude-conditional magnitude
No supporting evidence; within effect inconclusive	Any dynamic reading of the accumulated measures within Greece	Stage 5, no within estimate significant in the adverse direction, a
Model-dependent diagnostic	The absorption of Greece's residual by deprivation items	Stage 6, reverses from rank 3 to rank 25
Infeasible	Accumulated material resources: the source series begins in 2015 and no 2008 baseline exists	Stage 5, not a null result
Failed	The synthetic-control comparative design	Four of six pre-registered conditions; not shown

Inconclusive means the design could not have detected an effect of the relevant size. It is not evidence of absence.

Greek households report financial difficulty at a rate far above what the official relative-poverty rate predicts, and the distance is large, persistent and not a feature of a smooth European distribution. Three bodies of evidence make part of it more intelligible. None of them decomposes the gap, and no share of it is attributed to any of them.

The official measures are narrower than the experience. The broader AROPE measure closes about a fifth of the distance and its contribution is shrinking. The relative income line moved down with the Greek economy, which is what a relative measure is designed to do and which leaves it unable to register a decline affecting everyone at once. Together these are a real part of the answer and a minority of it.

What households report corresponds to concrete failure. Reported difficulty co-moves with arrears, inability to meet unexpected expenses, inadequate heating and severe deprivation, within countries as well as across them. This is corroboration from inside one survey instrument rather than independent validation, and it is not uniform across items.

Present conditions and accumulated exposure both carry information. Material resources, long-term unemployment and wage-adjusted affordability each predict hardship beyond income poverty. Accumulated unemployment, the duration of wage non-recovery and housing-cost deterioration each retain a cross-country association after their present-day counterparts are controlled.

What the evidence does not support

These limits are part of the conclusion, not qualifications appended to it.

- **No causal claim is made anywhere.** Every supported result is a cross-country association in a panel of twenty-seven countries. Nothing in this design identifies a causal effect.
- **There is no supporting dynamic evidence, and the within-country effect is inconclusive.** The accumulated measures differentiate countries; no within-country estimate supports a process unfolding within Greece, and the within tests are too imprecise to establish or rule one out. Those are two different statements and neither is "no effect".
- **Most non-supported tested constructs are unresolved, not excluded.** Six of nine current-level constructs are inconclusive under available power. Only a small number are excluded, and only at specific magnitudes.
- **A central absorption result reverses under a defensible alternative model.** Greece moves from third to twenty-fifth on unexplained hardship depending on whether a same-instrument predictor is admitted. Neither specification is definitive.
- **A broader negative reporting tendency is not ruled out.** The domain pattern suggests specificity; Greece is nonetheless close to worst in Europe on general life satisfaction. The EU-SILC series carries no pre-crisis observation. The separate ESS extension does reach back before 2008, and shows a Greek deficit that pre-dates the crisis — which makes a longstanding pattern more plausible, not less.
- **Most of the gap remains unexplained.** Of the 52.6-point average distance, the measurement account addresses a minority. The rest is not attributed here.

What follows for measurement

AUTHOR INTERPRETATION

POLICY IMPLICATIONS

The analysis shows repeatedly that no single official indicator captures what Greek households report. AROP misses concept and yardstick alike; AROPE adds breadth but a shrinking share of it; anchored poverty registers what the relative line cannot; and accumulated labour and housing exposure carries information none of the current-level measures hold.

A dashboard that reports these together would represent the situation better than any one of them alone. That is a recommendation about how poverty is reported, and it follows from what the analysis showed the single measures miss.

What may be concluded. POLICY RECOMMENDATION, NOT AN EMPIRICAL CONCLUSION: poverty dashboards should combine AROP, its real threshold, anchored poverty, AROPE/deprivation and accumulated labour and housing indicators.

Limitation. Presenting this as a finding. It follows from what the analysis showed AROP alone misses; it is not itself a result.

What would change this conclusion

Stating what would overturn a finding is more useful than asserting confidence in it. Four things would.

Independent corroboration from outside EU-SILC — administrative arrears data, utility disconnection records, or a separate survey instrument — would settle whether Stage 3's co-movement reflects shared substance or shared instrument, which is the question Stage 6 shows the current design cannot resolve. A longer or wider panel would convert several inconclusive results into genuine findings or genuine nulls; most of the six unresolved constructs are unresolved for want of power, not because the data speak against them. Household-level rather than country-level analysis would permit the within-country tests that this design cannot support, and would allow the dynamic question in Stage 5 to be asked properly. And respondent-level ESS access would put the pre-crisis wellbeing comparison on a firmer footing than the extension in Stage 7 can manage: that extension already indicates a Greek deficit predating the crisis, but it rests on approximate means reconstructed from published distributions, with no standard errors and no way to test anything.

Each of these is a route to a stronger answer than this report can give. It is worth saying plainly that this report's central finding is a well-documented gap with a partial account of its composition, and that the honest description of the remainder is that it is unexplained.

BACK MATTER

Protocol, data and reproduction

This section records how the analysis was governed. It is not on the main reading path, but every claim above depends on it being true.

How the analysis was constrained

Each construct in this report was specified, and the conditions it had to meet were written as executable code, before it was tested. Those conditions decide the outcomes reported in Stages 4 and 5; none of the verdicts is a judgement made after seeing the numbers.

The conditions are code rather than prose because prose admits interpretation exactly when interpretation is most tempting. A rule that runs cannot be reread more favourably once an inconvenient result arrives.

Where a result in this report was corrected during review, the correction and its reason are logged in the research record, which is generated from the same registers as this document.

Data

All quantitative inputs are Eurostat, principally EU-SILC, at country-year level for the EU27 over 2015–2024, with the constructed pre-2010 extension described in Stage 1 used for description only. The institutional trust figures come from the OECD's 2024 trust survey and were read from the primary release. Migration figures are national-level Greek data.

The literature cited in Stage 7 was verified against primary sources rather than secondary summaries: each reference was checked at its publisher or repository of record.

– **Sources, coverage, and how missing data is handled**

WHAT THE PANEL CONTAINS

The estimation panel is country-year, covering the EU27 across 2015–2024, drawn from Eurostat. The outcome and the deprivation items come from EU-SILC; the labour-market measures from the Labour Force Survey; prices and wages from the harmonised price and national accounts series; and purchasing-power figures from the PPP programme. The accumulated measures in Stage 5 reach back to 2008 or 2010 for their baselines, so their inputs extend earlier than the estimation window even though the outcome does not.

THE EU COMPARISON

Where a figure shows "the EU median" it is the median across the member states present in that year, recomputed per year rather than fixed to a base year's membership, and Greece is included in it. An excluded-Greece median would widen every gap shown here by construction, which would flatter the argument; the difference is small but it runs in the direction that favours the report's own case, so the conservative choice is the one taken.

MISSING VALUES

Nothing is imputed. Where a country-year lacks a value the observation is dropped from the model that needs it, which is why the row counts differ between specifications — and why Stage 6's two models are explicitly run on the intersection of their rows rather than on whatever each could manage alone. Series with structural breaks flagged by Eurostat are used as published; this project does not attempt its own break adjustment, and cross-vintage comparability before 2010 is a limitation noted in Stage 1 rather than a problem solved here.

TWO COVERAGE GAPS THAT SHAPED THE ANALYSIS

The purchasing-power consumption series begins in 2015, which is why the accumulated version of material resources could not be built at all rather than being built badly from a later baseline. And the wellbeing module begins in 2013, which is why the pre-crisis question in Stage 7 needed an entirely separate survey to address at all.

Scope, and what this design can carry

The unit of analysis is the country-year, and the panel holds twenty-seven countries over roughly a decade. Three consequences follow and none of them are incidental.

First, every result is an aggregate association. Nothing here describes an individual Greek household, and an aggregate relationship need not hold at the household level — the ecological inference problem is real and this design does not escape it. A finding that accumulated unemployment predicts national hardship rates does not establish that individuals who experienced unemployment report more hardship.

Second, the number of clusters is small. Twenty-seven is enough for descriptive comparison and marginal for inference, which is why the bootstrap gate exists and why so many constructs land as inconclusive. A larger panel would not merely tighten the estimates; it would convert a substantial part of this report from silence into evidence.

Third, the period is short relative to the process being studied. The accumulated measures reach back to 2008 or 2010, but the outcome series supports roughly a decade of variation, and within-country identification draws only on that. This is the direct cause of the imprecision in Stage 5, and it is why the dynamic question is left open rather than answered.

Provenance

Every claim, decision, correction and deviation in this project is recorded in the research record, along with which script produced each number and from which artifact. This report states its results; the record shows how each one was arrived at, and what was changed along the way.

– What is checked before this report is published

Every figure is checked mechanically before publication, and the checks that matter to a reader are these.

The numbers in each chart and the numbers in its table are compared by a checksum recomputed from the rendered table, so a chart cannot drift from the figures beneath it. Every view must contain real data — at least two positions on its axis and one finite value — because a chart that renders correctly while displaying nothing will otherwise pass unnoticed, as one did. No internal variable name may appear in a reader-facing label. Every figure must carry its question, its evidence status, a keyboard-reachable chart and a table fallback, and wide content must scroll inside its own container rather than pushing the page sideways.

Results that were registered as not reportable — principally the failed synthetic control's divergence chart — are checked for by name in every generated document, so they cannot reappear by accident. An optional input that is unavailable writes an explicit marker recording that it was skipped, so a stage that did not run can never be mistaken for one that ran and found nothing.

Reproduction

Every number in this report is produced from the published Eurostat, OECD and ESS sources by code, and the whole sequence can be rerun from those sources without manual steps. Which script produced each number, from which input, and under which decision, is recorded in the research record.